

JENKINS-TASK-04

❖ Importance of Jenkins Pipeline

A Jenkins pipeline is a **structured way to automate CI/CD tasks** from source code to quality checks and notifications.

★ Why Jenkins Pipeline is Important

1. Automation

- Automatically pulls code, checks quality, and sends alerts without manual effort.

2. Consistency

- Every build follows the same steps, reducing human errors.

3. Faster Feedback

- Developers get quick feedback on code quality and build status.

4. Early Bug Detection

- Integrating SonarQube helps detect bugs, vulnerabilities, and code smells early.

5. Improved Collaboration

- Slack notifications instantly inform the team about build results.

6. Scalability

- Pipelines can be extended later to include Maven, Docker, deployment, etc.

7. Traceability

- Each stage (clone, scan, notify) is logged and auditable.

◆ Importance of Tools Used

■ 1. Jenkins (CI Tool)

Role: Orchestrator of the pipeline

Why Jenkins is important:

- Automates the CI workflow
- Supports thousands of plugins
- Easy integration with Git, SonarQube, Slack

- Open-source and widely used in industry
-

2. Git & GitHub

Role: Source Code Management

Importance:

- Centralized code repository
 - Version control and history tracking
 - Enables collaboration among developers
 - Jenkins pulls code directly from GitHub
-

3. SonarQube

Role: Static Code Quality & Security Analysis

Why SonarQube is important:

- Detects bugs, vulnerabilities, and code smells
 - Enforces coding standards
 - Improves maintainability and security
 - Helps maintain high code quality before deployment
-

4. PostgreSQL

Role: SonarQube Backend Database

Importance:

- Officially supported database for SonarQube
 - Reliable and high-performance
 - Stores metrics, issues, quality gates, and history
 - Better stability compared to deprecated MySQL support
-

5. Slack

Role: Notification & Collaboration Tool

Why Slack integration is useful:

- Sends real-time build and scan alerts
 - Improves team communication
 - Reduces dependency on Jenkins UI
 - Helps in faster issue resolution
-

6. Docker (Used during troubleshooting)

Role: Alternative deployment option

Importance:

- Provides quick setup
 - Avoids dependency conflicts
 - Helps compare containerized vs traditional setup
 - Useful for learning and fallback deployment
-

7. Java (JDK 17)

Role: Runtime for Jenkins & SonarQube

Importance:

- Required for Jenkins and SonarQube execution
 - Correct Java version prevents compatibility issues
 - Ensures Elasticsearch and Sonar services run properly
-

Overall Architecture Importance

Developer → GitHub → Jenkins Pipeline



SonarQube Analysis



Slack Notification

- ✓ Ensures code quality
- ✓ Ensures automation
- ✓ Ensures team visibility

1. Setup Jenkins CICD pipeline using freestyle job using Docker containers using below code. <https://github.com/betawins/hiring-app.git> Stages:

Git Clone

SonarQube Integration

Maven Compilation

Nexus Artifactory

Slack Notification

Deploy On tomcat

launch one new instance for sonarqube server and inatall it with docker

Docker installed using yum install docker -y

amazon linux 2023 RPMs Livepatch repository				175 KB/s 29 KB	00:00
Dependencies resolved.					
Package	Architecture	Version	Repository	Size	
Installing:					
docker	x86_64	25.0.13-1.amzn2023.0.2	amazonlinux	66 M	
Installing dependencies:					
containerd	noarch	412.242.0-1.amzn2023	amazonlinux	50 K	
container-selinux	x86_64	2.1.5-1.amzn2023.0.1	amazonlinux	23 M	
iptables-libs	x86_64	1.8.6-3.amzn2023.0.2	amazonlinux	401 K	
iptables-nft	x86_64	1.8.6-3.amzn2023.0.2	amazonlinux	183 K	
libnftnl	x86_64	3.0-1.amzn2023.0.1	amazonlinux	75 K	
libnftables-common	x86_64	1.0.5-2.amzn2023.0.2	amazonlinux	58 K	
libnftables	x86_64	1.0.5-2.amzn2023.0.2	amazonlinux	30 K	

dnf install java-17-amazon-corretto -y

[root@ip-172-31-32-107 ~]# dnf install java-17-amazon-corretto -y				
Last metadata expiration check: 0:08:35 ago on Wed Dec 31 09:12:49 2023.				
Dependencies resolved.				
Package	Architecture	Version	Repository	Size
Installing:				
java-17-amazon-corretto	x86_64	1:17.0.17a10-1.amzn2023.1	amazonlinux	219 k
Installing dependencies:				
alsa-lib	x86_64	1:2.7.2-1.amzn2023.0.2	amazonlinux	504 k
attr	x86_64	1:15.0-4.amzn2023.0.2	amazonlinux	717 k
dejavu-mono-fonts	noarch	2.37-16.amzn2023.0.2	amazonlinux	1.3 M
dejavu-mono-misc-fonts	noarch	2.37-16.amzn2023.0.2	amazonlinux	447 M
dejavu-serif-fonts	noarch	2.37-16.amzn2023.0.2	amazonlinux	1.0 M
fontconfig	x86_64	2.13.94-2.amzn2023.0.2	amazonlinux	275 k
fontconfig-devel	x86_64	1:12.0.15-12.amzn2023.0.2	amazonlinux	9.5 k
fonttype	x86_64	2.13.2-5.amzn2023.0.1	amazonlinux	423 k
glibc	x86_64	5.2.15-9.amzn2023.0.2	amazonlinux	48 k
google-misc-fonts-common	noarch	20240401-1.amzn2023.0.2	amazonlinux	17 k
google-misc-misc-vf-fonts	noarch	20240401-1.amzn2023.0.2	amazonlinux	183 k
graphviz	x86_64	1.9.14-7.amzn2023.0.2	amazonlinux	97 k
harfbuzz	x86_64	7.0.0-2.amzn2023.0.2	amazonlinux	873 k
java-17-amazon-corretto-headless	x86_64	1:17.0.17a10-1.amzn2023.1	amazonlinux	91 M
java-common	x86_64	6.0.0-9.amzn2023.0.4	amazonlinux	12 k
langpacks-core-font-an	noarch	3.0-21.amzn2023.0.1	amazonlinux	10 k
libICE	x86_64	1.1.1-3.amzn2023.0.1	amazonlinux	76 k
libSM	x86_64	1.2.4-3.amzn2023.0.1	amazonlinux	45 k
libX11	x86_64	1.8.10-2.amzn2023.0.1	amazonlinux	859 k
libX11-common	noarch	1.0.10-2.amzn2023.0.1	amazonlinux	147 k
libXext	x86_64	1.0.11-6.amzn2023.0.1	amazonlinux	23 k

I am inside **Server-2 (SonarQube_with_docker)**

I already installed Docker

I added kernel values:

- vm.max_map_count = 524288
- fs.file-max = 131072

- sysctl -p output confirmed changes

After entering following commands

- echo 'vm.max_map_count=524288' | sudo tee -a /etc/sysctl.conf
- echo 'fs.file-max=131072' | sudo tee -a /etc/sysctl.conf
- sudo sysctl -p

```
Complete!
[root@ip-172-31-82-107 ~]# sudo sysctl -p
vm.max_map_count=524288
fs.file-max=131072
[root@ip-172-31-82-107 ~]# sysctl -p
vm.max_map_count=524288
fs.file-max=131072
[root@ip-172-31-82-107 ~]#
```

Start docker

- sudo systemctl start docker
- sudo systemctl enable docker
- sudo systemctl status docker

```
[root@ip-172-31-82-107 ~]# sudo systemctl start docker
sudo systemctl enable docker
sudo systemctl status docker
Created symlink /etc/systemd/system/multi-user.target.wants/docker.service - /usr/lib/systemd/system/docker.service.
* docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; disabled; preset: disabled)
   Active: active (running) since Wed 2025-12-31 09:32:26 UTC; 40s ago
   TriggeredBy: * docker.socket
   Docs:
     - https://docs.docker.com
   Main PID: 28732 (dockerd)
   Tasks: 10
   Memory: 80.1M
   CPU: 395ms
   CGroup: /system.slice/docker.service
           └─ 28732 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock --default-ulimit nofile=12768:65536

Dec 31 09:32:25 ip-172-31-82-107.ec2.internal systemd[1]: Starting docker.service - Docker Application Container Engine...
Dec 31 09:32:25 ip-172-31-82-107.ec2.internal dockerd[28732]: time="2025-12-31T09:32:25.735235412Z" level=info msg="Starting up"
Dec 31 09:32:25 ip-172-31-82-107.ec2.internal dockerd[28732]: time="2025-12-31T09:32:25.801118482Z" level=info msg="Loading containers: start."
Dec 31 09:32:26 ip-172-31-82-107.ec2.internal dockerd[28732]: time="2025-12-31T09:32:26.378276388Z" level=info msg="Loading containers: done."
Dec 31 09:32:26 ip-172-31-82-107.ec2.internal dockerd[28732]: time="2025-12-31T09:32:26.406228984Z" level=info msg="Docker daemon" commit="165516e containerd=mapshotter=false storage-driver=overlay2 version=25.0.13
Dec 31 09:32:26 ip-172-31-82-107.ec2.internal dockerd[28732]: time="2025-12-31T09:32:26.406402145Z" level=info msg="Daemon has completed initialization"
Dec 31 09:32:26 ip-172-31-82-107.ec2.internal dockerd[28732]: time="2025-12-31T09:32:26.463049140Z" level=info msg="API listen on /run/docker.sock"
```

Install and Run SonarQube Container

- docker run -d --name sonarqube \
- -p 9000:9000 \
- sonarqube:its

```
Dec 31 09:32:25 ip-172-31-82-107.ec2.internal systemd[1]: Started docker.service - Docker Application Container Engine.
[root@ip-172-31-82-107 ~]# docker run -d --name sonarqube \
  -p 9000:9000 \
  sonarqube:its
Unable to find image 'sonarqube:its' locally
its: Pulling from library/sonarqube
60d98d907669: Pull complete
c24a8b9e552f: Pull complete
f392b3e4e58b: Pull complete
1d7735c481ad: Pull complete
8d8a1ad7025: Pull complete
6b77b3a980a1: Pull complete
c7ad1fe1e07: Pull complete
14f4d70e454: Pull complete
Digest: sha256:f708979ab31d2d08f5a3e2dc73a31ee011efc8cf28845082c17c55d45df9df5
Status: Downloaded newer image for sonarqube:its
6b794e85f11c0308ba14c3ba32afdf9090ba0c6da06876c43d2029ac69ad8951
[root@ip-172-31-82-107 ~]#
```

Given security permission to port 9000 for soanrqube

Inbound rules (2)									
Search Manage tags Edit inbound rules									
☐	Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	Description	
☐	-	sgr-0eb07fffa7e3456b9	IPv4	SSH	TCP	22	0.0.0.0/0	-	
☐	-	sgr-07d541825668cb3eb	IPv4	Custom TCP	TCP	9000	0.0.0.0/0	sonarqube	

Sonarqube is now up and running

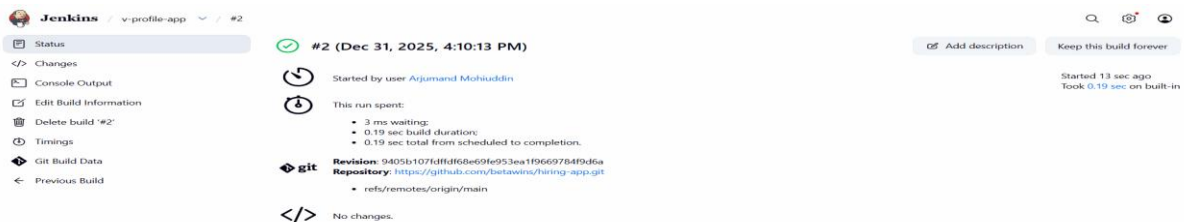


Changed sonar path in terminal. Path will be /opt/sonar-scanner/conf/sonar-scanner-properties



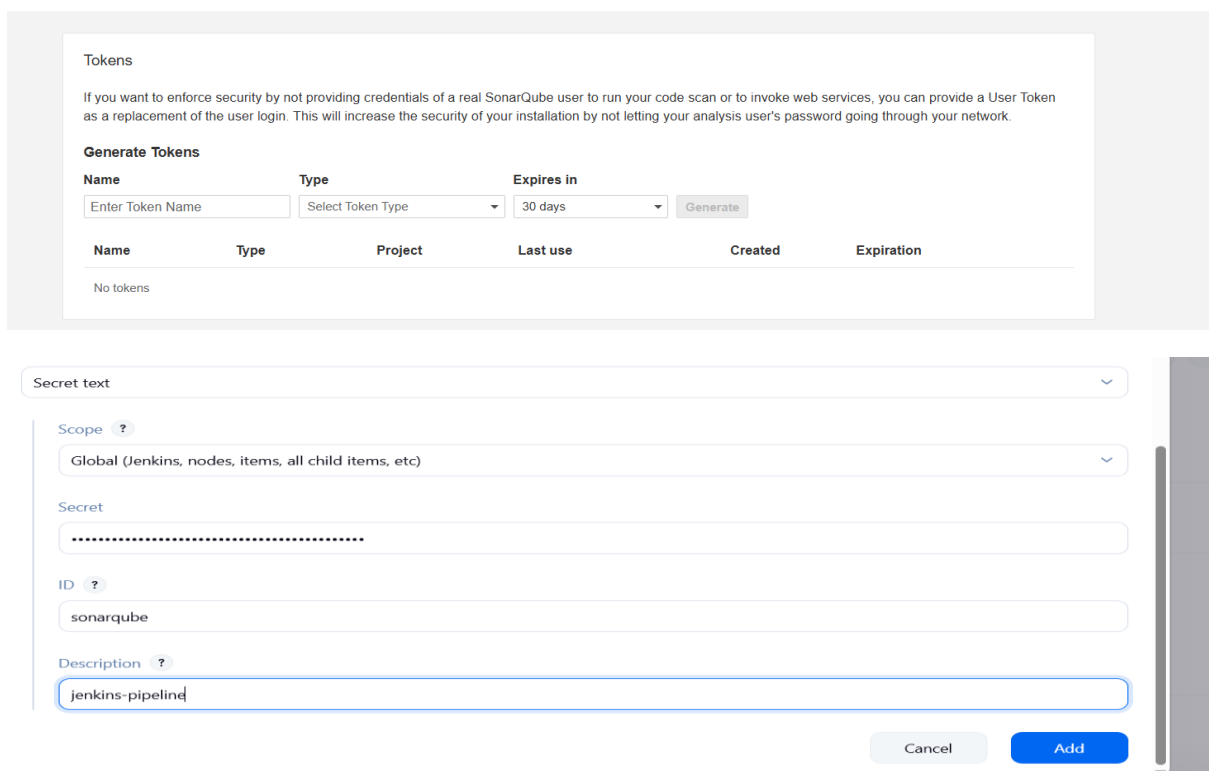
1. Clone the job using git repo URL.

- Create a new job with freestyle and choose git and add git repo



1. Sonarqube setup for the junkins pipeline

Create token and add in Jenkins



Install maven using commands

- `sudo apt update -y`
- `sudo apt install -y maven`

```
Installed:
apache-commons-cli-1.10.0-3.amzn2023.0.3.noarch
atinject-1.0.5-3.amzn2023.0.3.noarch
httpcomponents-client-4.5.13-4.amzn2023.0.4.noarch
java-17-amazon-corretto-devel-117.0.1710-1.amzn2023.1.x86_64
maven-13.8.4-3.amzn2023.0.5.noarch
maven-shared-utils-3.3.4-4.amzn2023.0.4.noarch
plexus-containers-component-annotations-2.1.0-9.amzn2023.0.4.noarch
publicsuffix-list-20240212-61.amzn2023.noarch

apache-commons-codec-1.15-6.amzn2023.0.3.noarch
cdi-spi-2.0.2-6.amzn2023.0.3.noarch
httpcomponents-core-4.4.13-6.amzn2023.0.3.noarch
jib-core-0.17.0-3.amzn2023.0.4.noarch
maven-animal-sniffer-1.18.4-3.amzn2023.0.5.noarch
maven-wagon-3.4.2-4.amzn2023.0.4.noarch
plexus-interpolation-1.26-10.amzn2023.0.4.noarch
sisu-1.0.3.4-9.amzn2023.0.4.noarch

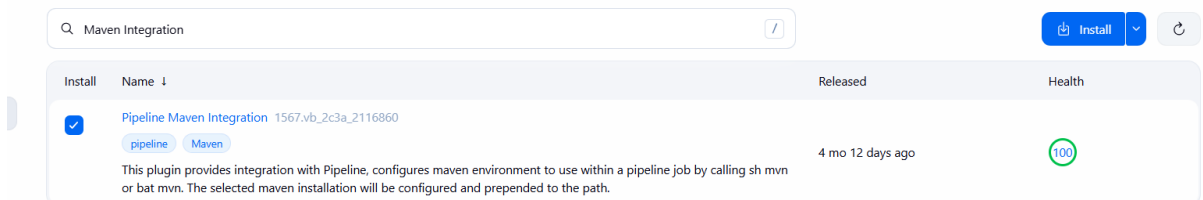
apache-commons-io-112.8.0-7.amzn2023.0.5.noarch
google-guava-1.2.3-4.amzn2023.0.6.noarch
httpcomponents-httpclient-4.5.13-6.amzn2023.0.3.noarch
jakarta-annotations-1.3.5-13.amzn2023.0.3.noarch
jansi-2.4.0-3.amzn2023.0.3.x86_64
jnr-constants-0.10.2-8.amzn2023.0.4.noarch
maven-resolver-1.11.7-3.amzn2023.0.4.noarch
plexus-cipher-1.8-3.amzn2023.0.3.noarch
plexus-sec-dispatcher-2.0-3.amzn2023.0.3.noarch
slf4j-1.7.32-3.amzn2023.0.4.noarch

apache-commons-lang3-3.10.0-1.amzn2023.0.1.noarch
guava-31.0.1-3.amzn2023.0.6.noarch
jansi-2.4.0-3.amzn2023.0.3.x86_64
jnr-constants-0.10.2-8.amzn2023.0.4.noarch
maven-resolver-1.11.7-3.amzn2023.0.4.noarch
plexus-classworlds-2.6.0-10.amzn2023.0.4.noarch
plexus-utils-3.3.0-9.amzn2023.0.4.noarch

Complete!
[ec2-user@ip-172-31-21-117 ~]$ mvn -version
Apache Maven 3.8.4 (Red Hat 3.8.4-3.amzn2023.0.5)
Maven home: /usr/share/maven
Java version: 17.0.17, vendor: Amazon.com Inc., runtime: /usr/lib/jvm/java-17-amazon-corretto.x86_64
Default locale: en, platform encoding: UTF-8
OS name: "linux", version: "4.12.18-62.121.amzn2023.x86_64", arch: "amd64", family: "unix"
[ec2-user@ip-172-31-21-117 ~]$ mvn
/home/ec2-user/
[ec2-user@ip-172-31-21-117 ~]$ ls
looking_backup.sh test.pom
[ec2-user@ip-172-31-21-117 ~]$ ||
```

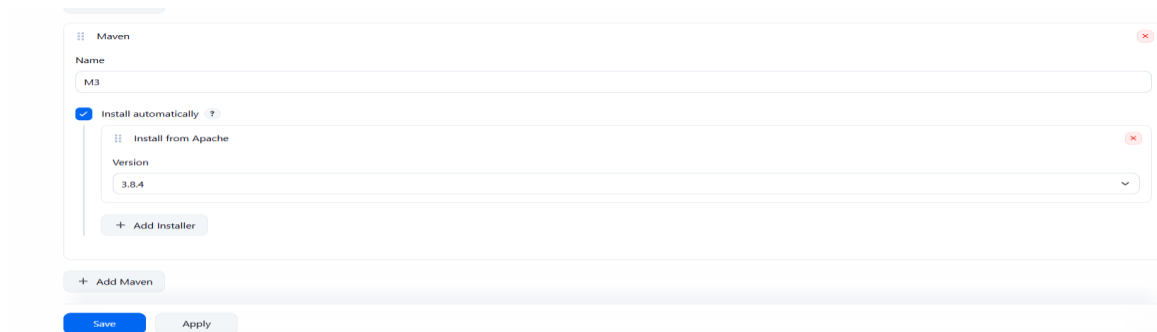
Install pugin in Jenkins GUI

- Go to manage Jenkins < plugins < install maven integration



2. MVAEN setup in Jenkins server

- Then go to tools add maven details as shown in the image below



Configure job and add these details in invoke top-level maven targets

sq_a_9ce0f3955c54bc86fcd40170f60049be952a372

Invoke top-level Maven targets ?

Maven Version

M3

Goals

clean package

Advanced

+ Add build step

Maven Version

M3

Goals

clean package

Advanced

Execute SonarQube Scanner

JDK ?

JDK to be used for this SonarQube analysis

(Inherit From Job)

Path to project properties ?

Analysis properties

sonar.projectKey=hiring-app
sonar.projectName=hiring-app
sonar.projectVersion=1.0
sonar.sources=src/main/webapp
sonar.sourceEncoding=UTF-8
sonar.host.url=http://3.88.130.153:9000
sonar.login=sqa_9ce0f3955c54bc86fcd40170f60049beeb952a372

Jenkins v-profile-app #8

Status #8 (Dec 31, 2025, 5:39:42 PM)

Changes

Console Output

Edit Build Information

Delete build '#8'

Timings

Git Build Data

Previous Build

Started by user Arjumand Mohiuddin

This run spent:

- 7 ms waiting;
- 14 sec build duration;
- 14 sec total from scheduled to completion.

Revision: 9405b1071dfdf68e69fe953ea1f9669784f9d6a
Repository: <https://github.com/betawins/hiring-app.git>

- refs/remotes/origin/main

No changes.

Add description

Keep this build forever

Started 23 sec ago
Took 14 sec on built-in

Code is transferred to sonarqube and code is passed

Filters

Quality Gate

Passed

Failed

Reliability (Bugs)

A rating

B rating

C rating

1 project(s)

Perspective: Overall Status Sort by: Name

hiring-app Passed

Last analysis: 2 minutes ago

Bugs 3

Vulnerabilities 0

Hotspots Reviewed

Code Smells 0

Coverage

Duplications 0.0%

Lines 11

XML, JSP

1 of 1 shown

Validation: **Maven**

Target directory is created

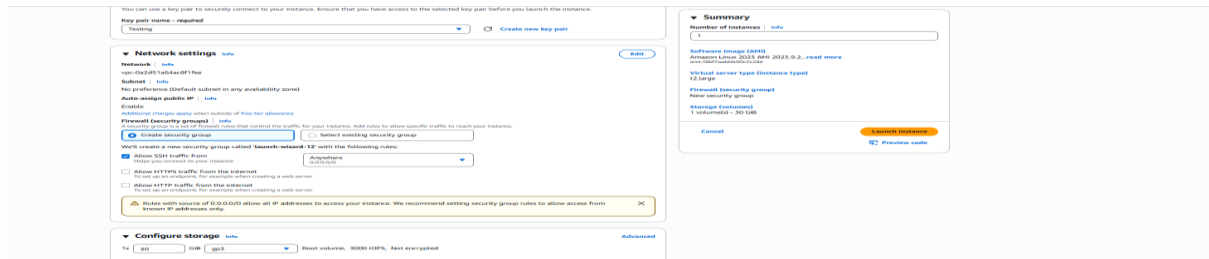
```
East login: Wed Dec 31 16:04:42 2025 from 19.206.107.29
[ec2-user@ip-172-31-21-117 ~]$ cd /var/lib/jenkins/workspace/
[ec2-user@ip-172-31-21-117 workspace]$ ls
sample_job sample_job2 first_job github-private-repo github-private-repo1mp postgresql-job prediocioal_job v-profile-app
[ec2-user@ip-172-31-21-117 workspace]$ cd v-profile-app/
[ec2-user@ip-172-31-21-117 v-profile-app]$ ls
Dockerfile Jenkinsfile README.md 'Untitled Diagram.drawio' jenkinsfile-cicd pom.xml src target
[ec2-user@ip-172-31-21-117 v-profile-app]$
```


Hiring.war file created

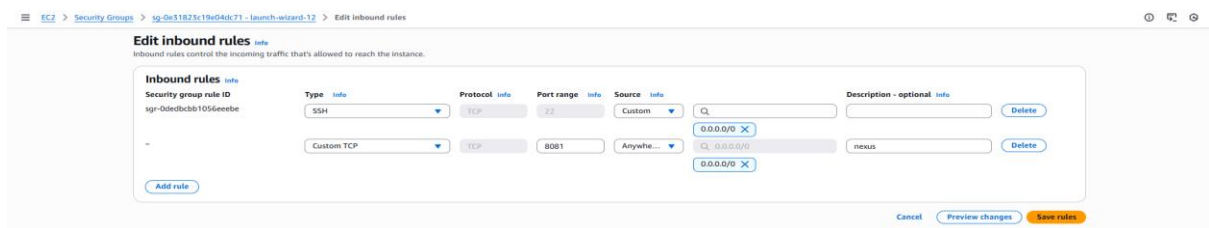
```
[ec2-user@ip-172-31-21-117 v-profile-app]$ cd target/
[ec2-user@ip-172-31-21-117 target]$ ls
hiring hiring.war maven-archiver
[ec2-user@ip-172-31-21-117 target]$ ||
```

3. Nexus setup for jenkins pipeline

Launch one more server for nexus



I added 8081 to be allowed in SG inbound role as nexus works on 8081.



Install Java 17 (Required for Nexus)

SSH into server & run:

- `sudo yum update -y`
- `sudo yum install java-17-amazon-corretto-headless -y`
- `java -version`

```
[ec2-user@ip-172-31-84-50 ~]$ sudo yum update -y
sudo yum install java-17-amazon-corretto-headless -y
java -version
Amazon Linux 2023 Kernel livepatch repository
Last metadata expiration check: 0:00:03 ago on Wed Dec 31 18:05:13 2025.
Dependencies resolved.
Nothing to do.
Complete!
Last metadata expiration check: 0:00:02 ago on Wed Dec 31 18:05:13 2025.
Dependencies resolved.

Package Architecture Version Repository Size
Installing:
java-17-amazon-corretto-headless x86_64 1:17.0.17+10-1, amzn2023.1 amazonlinux 91 M
Installing dependencies:
alsa-lib x86_64 1.2.7.2-1, amzn2023.0.2 amazonlinux 504 K
alsa-ucm-conf x86_64 1.18.0-4, amzn2023.0.3 amazonlinux 715 K
alsa-utils x86_64 2.37-16, amzn2023.0.2 amazonlinux 1.3 M
alsa-utils-libs x86_64 2.37-16, amzn2023.0.2 amazonlinux 467 K
alsa-utils-libs-libs x86_64 2.37-16, amzn2023.0.2 amazonlinux 1.0 M
alsa-utils-libs-libs-devel x86_64 2.13.94-2, amzn2023.0.2 amazonlinux 279 K
```

```
[ec2-user@ip-172-31-84-50 ~]$ cd /opt/
[ec2-user@ip-172-31-84-50 opt]$ sudo wget https://download.sonatype.com/nexus/3/latest-unix.tar.gz
sudo tar -xvzf latest-unix.tar.gz
sudo mv nexus-3* nexus
sudo rm latest-unix.tar.gz
--2025-12-31 18:11:52-- https://download.sonatype.com/nexus/3/latest-unix.tar.gz
Resolving download.sonatype.com (download.sonatype.com)... 54.241.7.25, 54.215.226.116
```

Correct Nexus Download Command (Latest version)

- Let's install **Nexus Repository OSS 3.68.1-02** for Linux
- `cd /opt`
- `sudo wget https://download.sonatype.com/nexus/3/nexus-3.68.1-02-unix.tar.gz`
- `sudo tar -xvf nexus-3.68.1-02-unix.tar.gz`

- ```

nausa-6.48.1-02/system/crg/somatype/nausa/nausa-test-log-cleanup-3.48.1-02.jar
nausa-6.48.1-02/system/crg/somatype/osx/osx/osx-service-api-1.3.0/osx-service-api-1.3.0.jar
nausa-6.48.1-02/system/crg/somatype/osx/osx/osx-service-client-1.3.0/osx-service-client-1.3.0.jar
nausa-6.48.1-02/system/crg/somatype/sim/sim-odata4/0.0.7/sim-odata4-0.0.7.jar
nausa-6.48.1-02/system/crg/somatype/web-library/1.7.0/web-web-library-1.7.0.jar
nausa-6.48.1-02/system/crg/spdx/spdx-jackson-rcv/1.1.7/spdx-jackson-rcv-1.1.7.jar
nausa-6.48.1-02/system/crg/com/cml/cml/1.0.0/cml-1.0.0.jar
nausa-6.48.1-02/system/crg/tubiana/t/1.3/t-1.3.jar
nausa-6.48.1-02/system/crg/yaml/makeyaml/2.2/makeyaml-2.2.jar
nausa-6.48.1-02/system/crg/sonoma/java/alphanumeric-comparator/1.4.1/alphanumeric-comparator-1.4.1.jar
nausa-6.48.1-02/system/crg/etcd/etcd.jar
nausa-6.48.1-02/system/uk/gov/ids/ids/synout-over-ids/1.0.2/synout-over-ids-1.0.2.jar
nausa-6.48.1-02/system/uk/gov/ids/ids/synout-min-1.4.0/synout-min-1.4.0.jar
nausa-work/nausa/clean_cache
nausa-work/nausa/log_placeholder
nausa-work/nausa/nausa-gi-ko-helper
[ec2-user@ip-172-31-84-50 opt]# ls
nausa-work-6.48.1-02.jar nausa-work
[ec2-user@ip-172-31-84-50 opt]# |

```

- `sudo useradd nexus`
- `sudo chown -R nexus:nexus /opt/nexus`
- `sudo chown -R nexus:nexus /opt/sonatype-work`

```
sonatype-work@nexus3: /opt/sonatype-work$ ls
[ec2-user@ip-172-31-84-50 ~]$ ls
aws nexus nexus-3.68-1-02-unix.tar.gz sonatype-work
[ec2-user@ip-172-31-84-50 ~]$ sudo su -
[root@ip-172-31-84-50 ~]# sudo unxzad nexus
[root@ip-172-31-84-50 ~]# sudo chown -R nexus:nexus /opt/nexus
sudo chown -R nexus:nexus /opt/sonatype-work
[root@ip-172-31-84-50 ~]# sudo vi /opt/nexus/bin/nexus.rc
[root@ip-172-31-84-50 ~]# |||
```

- Edit the file:
- `sudo vi /opt/nexus/bin/nexus.rc`
- Add this line at bottom:
- `run_as_user="nexus"`

```
run_as_user="nexus"|
```

- `sudo vi /etc/systemd/system/nexus.service`

```
[root@172-31-84-50 ~]# cd /etc/systemd/system
[root@172-31-84-50 systemd]# ls
basic.target.wants dbus-org.freedesktop.homed.service getty.target.wants nfs-idmapd.service.requires rpo-gssd.service.requires sysinit.target.wants
chronyd.service.wants dbus-org.freedesktop.networkd.service multi-user.target.wants nfs-mountd.service.requires rpo-statd-notify.service.requires systemd.wants
cronind.target.wants dbus-org.freedesktop.resolve.service network-online.target.wants nfs-server.service.requires rpo-statd.service.requires systemd-homed.service.wants
crond-all.target dbus.target nfs-blkio.service.requires remote-fs.target.wants sockets.target.wants systemd.target.wants
[root@172-31-84-50 systemd]# vi /etc/systemd
```

```
[Unit]
Description=nexus service
After=network.target

[Service]
Type=forking
LimitNOFILE=65536
User=nexus
Group=nexus
ExecStart=/opt/nexus/bin/nexus start
ExecStop=/opt/nexus/bin/nexus stop
Restart=on-abort

[Install]
WantedBy=multi-user.target
```

## Run:

- `sudo systemctl daemon-reload`

- sudo systemctl enable nexus
- sudo systemctl start nexus
- sudo systemctl status nexus

## ERROR:

### ❖ Nexus was unable to run if I check status

Trouble shoot:

- 👉 Nexus is still using Java 17
- 👉 That version is **NOT supported** — must be Java 8 or Java 11

So, we must:

- 1 Install Java 11
- 2 Configure Nexus to use **Java 11** instead of default Java 17

```
[root@ip-172-31-84-50 ~]# systemctl daemon-reload
[root@ip-172-31-84-50 ~]# systemctl enable nexus
[root@ip-172-31-84-50 ~]# systemctl start nexus
Job for nexus.service failed because the control process exited with error code.
See "systemctl status nexus.service" and "journalctl -xeu nexus.service" for details.
[root@ip-172-31-84-50 ~]# systemctl status nexus
* nexus.service - nexus.service
 Loaded: loaded (/etc/systemd/system/nexus.service; enabled; preset: disabled)
 Active: failed (Result: exit-code) since Wed 2025-12-31 18:40:28 UTC; 13s ago
 Process: 28559 ExecStart=/opt/nexus/bin/nexus start (code=exited, status=83)
 CPU: 43msa
Dec 31 18:40:27 ip-172-31-84-50.ec2.internal systemd[1]: Starting nexus.service - nexus.service.
Dec 31 18:40:28 ip-172-31-84-50.ec2.internal nexus[28559]: No suitable Java Virtual Machine could be found on your system.
Dec 31 18:40:28 ip-172-31-84-50.ec2.internal nexus[28559]: The version of the JVM must be at least 1.8 and at most 11.
Dec 31 18:40:28 ip-172-31-84-50.ec2.internal nexus[28559]: Please define INSTALL_JAVA_HOME to point to a suitable JVM.
Dec 31 18:40:28 ip-172-31-84-50.ec2.internal systemd[1]: nexus.service: Control process exited, code=exited, status=83/n/a
Dec 31 18:40:28 ip-172-31-84-50.ec2.internal systemd[1]: nexus.service: Failed with result 'exit-code'.
Dec 31 18:40:28 ip-172-31-84-50.ec2.internal systemd[1]: Failed to start nexus.service - nexus.service.
[root@ip-172-31-84-50 ~]# java -version
openjdk version "17.0.17" 2025-10-21 LTS
OpenJDK Runtime Environment Corretto-17.0.17.10.1 (build 17.0.17+10-LTS)
OpenJDK 64-Bit Server VM Corretto-17.0.17.10.1 (build 17.0.17+10-LTS, mixed mode, sharing)
```

## Install Java 11 (if not already installed)

- sudo yum install -y java-11-amazon-corretto-headless
- Verify installation:
- ls -l /usr/lib/jvm/

```
java-11-amazon-corretto-headless x86_64 1:11.0.29-b7-1.amzn2023 amazonlinux 01 M
Transaction Summary
Install 1 Package
Total download size: 91 M
Installed size: 226 M
Downloading Packages:
java-11-amazon-corretto-headless-11.0.29-b7-1.amzn2023.x86_64.rpm 61 MB/s | 91 MB 00:01
Total
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
Installing : java-11-amazon-corretto-headless-11.0.29-b7-1.amzn2023.x86_64 1/1
Running scriptlet: java-11-amazon-corretto-headless-11.0.29-b7-1.amzn2023.x86_64 1/1
Verifying : java-11-amazon-corretto-headless-11.0.29-b7-1.amzn2023.x86_64 1/1
Installed:
java-11-amazon-corretto-headless-11.0.29-b7-1.amzn2023.x86_64
Complete!
[root@ip-172-31-84-50 ~]# ls -l /usr/lib/jvm/
total 4
drwxr-xr-x. 7 root root 79 Dec 31 18:44 java-11-amazon-corretto.x86_64
lrwxrwxrwx. 1 root root 41 Dec 31 18:05 java-17-amazon-corretto -> /etc/alternatives/java-17-amazon-corretto
drwxr-xr-x. 7 root root 79 Dec 31 18:05 java-17-amazon-corretto.x86_64
lrwxrwxrwx. 1 root root 21 Dec 31 18:05 jre -> /etc/alternatives/jre
lrwxrwxrwx. 1 root root 24 Dec 31 18:05 jre-11 -> /etc/alternatives/jre_11
lrwxrwxrwx. 1 root root 25 Dec 31 18:05 jre-11-openjdk -> /etc/alternatives/jre_11-openjdk
lrwxrwxrwx. 1 root root 24 Dec 31 18:05 jre-17 -> /etc/alternatives/jre_17
lrwxrwxrwx. 1 root root 25 Dec 31 18:05 jre-17-openjdk -> /etc/alternatives/jre_17-openjdk
lrwxrwxrwx. 1 root root 29 Dec 31 18:05 jre-openjdk -> /etc/alternatives/jre-openjdk
[root@ip-172-31-84-50 ~]#
```

## Edit Systemd service file

- sudo vi /etc/systemd/system/nexus.service
- Ensure this exists inside [Service] section:

- Environment="JAVA\_HOME=/usr/lib/jvm/java-11-amazon-corretto"
- ExecStart=/opt/nexus/bin/nexus start
- ExecStop=/opt/nexus/bin/nexus stop

```
[Unit]
Description=nexus service
After=network.target

[Service]
Type=forking
LimitNOFILE=65536
User=nexus
Group=nexus
Environment=JAVA_HOME=/usr/lib/jvm/java-11-amazon-corretto
ExecStart=/opt/nexus/bin/nexus start
ExecStop=/opt/nexus/bin/nexus stop
Restart=on-abort

[Install]
WantedBy=multi-user.target
```

## Edit nexus.rc file

- sudo vi /opt/nexus/bin/nexus.rc
- Add or update:
- run\_as\_user="nexus"
- INSTALL4J\_JAVA\_HOME=/usr/lib/jvm/java-11-amazon-corretto

```
run_as user="nexus"
INSTALL4J_JAVA_HOME=/usr/lib/jvm/java-11-amazon-corretto
```

```
run_as user="nexus"
INSTALL4J_JAVA_HOME=/usr/lib/jvm/java-1.8.0-amazon-corretto
```

```
[Unit]
Description=nexus service
After=network.target

[Service]
Type=forking
LimitNOFILE=65536
User=nexus
Group=nexus
Environment=JAVA_HOME=/usr/lib/jvm/java-1.8.0-amazon-corretto
ExecStart=/opt/nexus/bin/nexus start
ExecStop=/opt/nexus/bin/nexus stop
Restart=on-abort

[Install]
WantedBy=multi-user.target
```

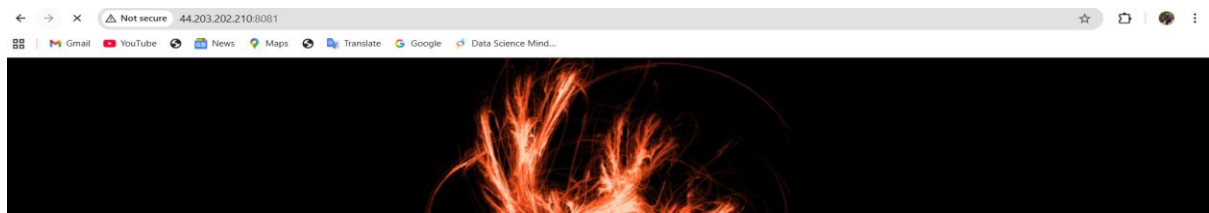
## Conclusion of the Error:

- 1 Nexus 3 supports only **Java 8** → Other Java versions cause startup failure.
- 2 Set correct JAVA\_HOME inside /etc/systemd/system/nexus.service using INSTALL4J\_JAVA\_HOME.
- 3 Ensure Java path is valid: /usr/lib/jvm/java-1.8.0-amazon-corretto.
- 4 Reload services after config changes: sudo systemctl daemon-reload && sudo systemctl restart nexus.
- 5 Check startup issues in logs: sudo journalctl -u nexus.service -n 50 or /opt/sonatype-

```
sudo: 3 incorrect password attempts
[nexus@ip-172-31-84-50 ~]$
[nexus@ip-172-31-84-50 ~]$ exit
logout
[root@ip-172-31-84-50 system]# systemctl daemon-reload
[root@ip-172-31-84-50 system]# systemctl restart nexus
[root@ip-172-31-84-50 system]# systemctl status nexus -l
● nexus.service - nexus service
 Loaded: loaded (/etc/systemd/system/nexus.service; enabled; preset: disabled)
 Active: active (running) since Wed 2025-12-31 19:35:29 UTC; 10s ago
 Process: 33562 ExecStart=/opt/nexus/bin/nexus start (code=exited, status=0/SUCCESS)
 Main PID: 33773 (java)
 Tasks: 41 (limit: 9492)
 Memory: 800.4M
 CPU: 19.397s
 CGroup: /system.slice/nexus.service
 └─33773 /usr/lib/jvm/java/bin/java -server -Dinstall4j.jvmDir=/usr/lib/jvm/java -Dexe4j.moduleName=/opt/nexus/bin/nexus -XX:+UnlockDiagnosticVMOptions -Dinstall4j.launcherId=245 -Dinstall4j.swt=false -Di4

Dec 31 19:35:29 ip-172-31-84-50.ec2.internal systemd[1]: Starting nexus.service - nexus service...
Dec 31 19:35:29 ip-172-31-84-50.ec2.internal nexus[33562]: Starting nexus
Dec 31 19:35:29 ip-172-31-84-50.ec2.internal systemd[1]: Started nexus.service - nexus service.
```

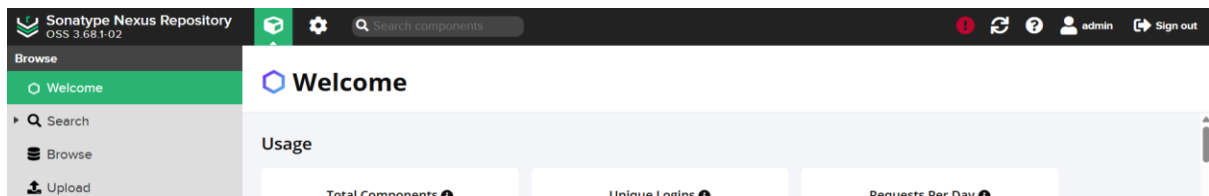
Nexus is opened



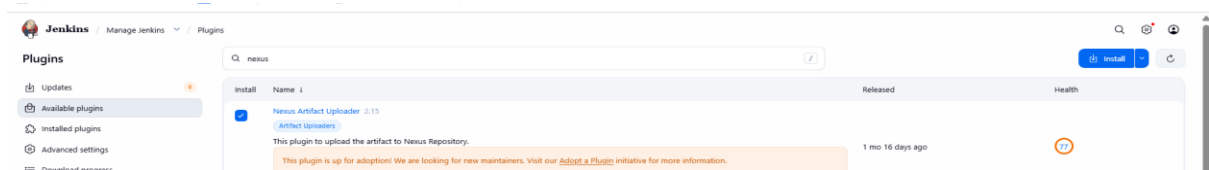
## Default Credentials:

- Username: admin
- Password: Stored inside this file:  
`sudo cat /opt/sonatype-work/nexus3/admin.password`

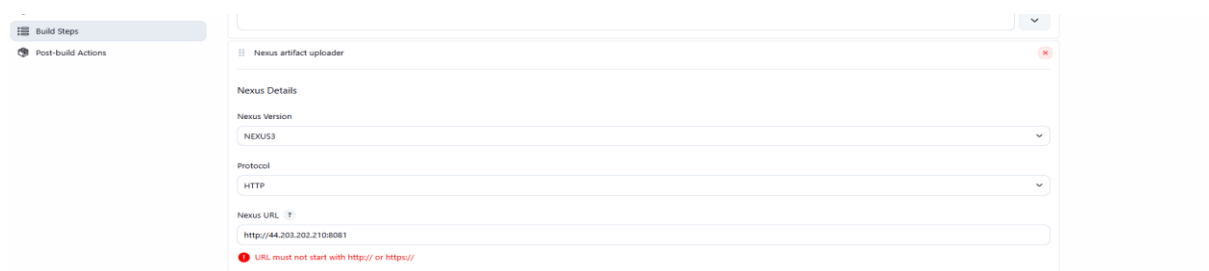
Setup new password then we are able to login to nexus.



## Install nexus plugin in Jenkins GUI



Nexus setup in GUI Jenkins add details and create repo in nexus rest of the details we will find in pom.xml, github repo



## Set credentials

Nexus artifact uploader

Jenkins Credentials Provider: Jenkins

Kind  
Username with password

Scope ?  
Global (Jenkins, nodes, items, all child items, etc)

Username ?  
admin

☐ Treat username as secret ?

Password ?  
\*\*\*\*\*

ID ?  
Nexus\_server

Description ?  
Nexus\_server

Cancel Add

Version  
0.1

Repository ?  
hiring\_Maven\_Webapp

Artifacts

Artifact

ArtifactId  
hiring

Type ?  
war

Classifier ?

File ?  
target/hiring.war

Group Id and version we can add from pom.xml from repo

```
<groupId>in.javahome</groupId>
<artifactId>hiring</artifactId>
<packaging>war</packaging>
<version>0.1</version>
```

Create repository in nexus GUI server

Administration

Repository

Repositories

Blob Stores

Repositories Manage repositories

Create repository

| Name ↑        | Type  | Format | Blob Store | Status                    | URL  | Health check | Firewall Re... |
|---------------|-------|--------|------------|---------------------------|------|--------------|----------------|
| maven-central | proxy | maven2 | default    | Online - Ready to Conn... | copy | Analyze      |                |
| maven public  | group | maven? | default    | Online                    |      |              |                |

Select maven 2 hosted

Administration

Repository

Repositories

Blob Stores

Proprietary Repositories

Content Selectors

Cleanup Policies

Routing Rules

Repositories / Select Recipe

Recipe ↑

gitifs (hosted)

go (group)

go (proxy)

helm (hosted)

helm (proxy)

maven2 (group)

maven2 (hosted)

maven2 (proxy)

npm (group)

npm (hosted)

What type of artifacts does this repository store?

Release

Layout policy:

Validate that all paths are maven artifact or metadata paths

Permissive

Content Disposition:

Add Content-Disposition header as 'Attachment' to disable some content from being inline in a browser.

Inline

Storage

Blob store:

Blob store used to store repository contents

default

Strict Content Type Validation:

☒ Validate that all content uploaded to this repository is of a MIME type appropriate for the repository format

Hosted

Deployment policy:

Controls if deployments of and updates to artifacts are allowed

Allow redeploy

Job is executed successfully

Jenkins / v-profile-app / #9

Status

Changes

Console Output

Edit Build Information

Delete build #9

Timings

Git Build Data

Previous Build

#9 (Dec 31, 2025, 8:19:00 PM)

Started by user Arjumand Mohiuddin

This run spent:

- 4 ms waiting;
- 12 sec build duration;
- 12 sec total from scheduled to completion.

Revision: 9405b107f0f0f6a6d9f953ea1f9669784f9dda

Repository: <https://github.com/betawins/hiring-app.git>

ref/remotes/origin/main

No changes.

Add description

Keep this build forever

Started 24 sec ago

Took 12 sec on built-in

Validation: this how repo is created in nexus

Browse

Welcome

Search

Custom

Maven

NuGet

Browse

Upload

Browse / hiring\_Maven\_Webapp

Upload component

HTML View

in

javahome

hiring

01

hiring-01.war

hiring-01.war.md5

hiring-01.war.sha1

## 4. Slack Notification:

Go to tools in slack and click tools

Slack

Workspace ?

techiehorizon

Credential ?

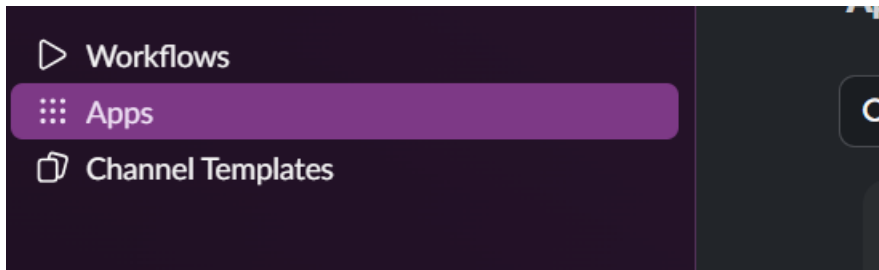
slack-notification

+ Add

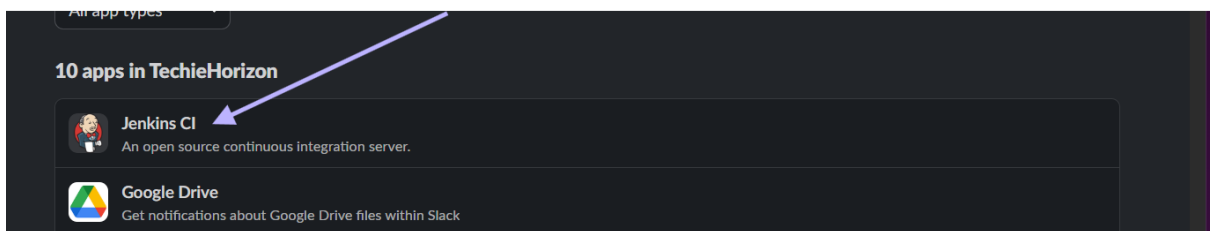
Default channel / member id ?



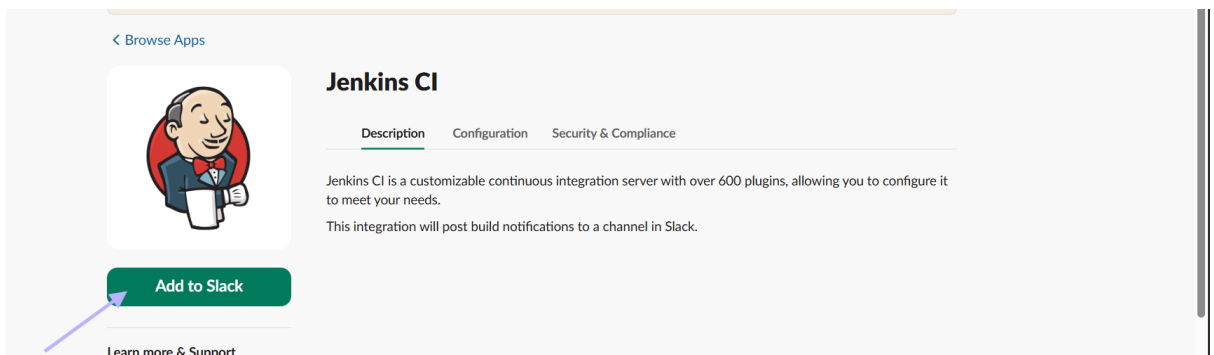
Select apps



Select Jenkins CI

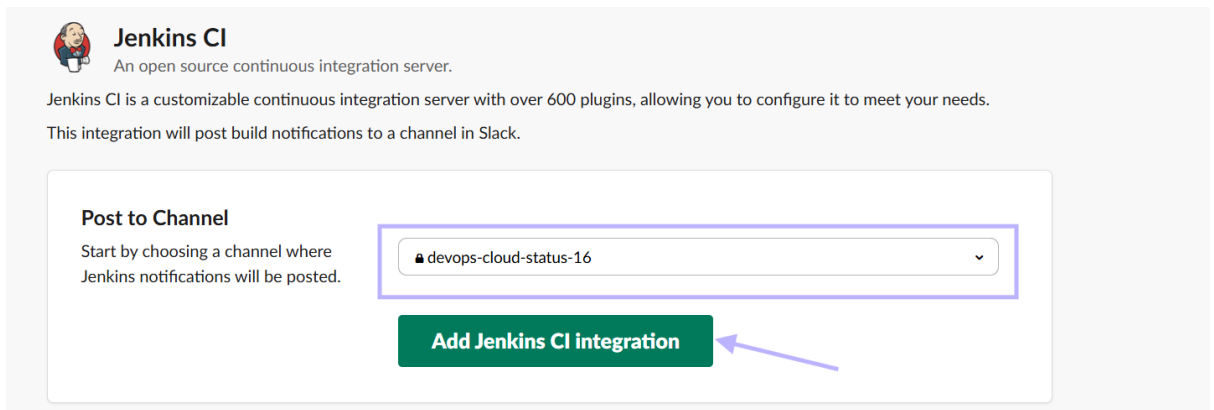


ADD Jenkins CI notifications



Select the channel





**Jenkins CI**  
An open source continuous integration server.

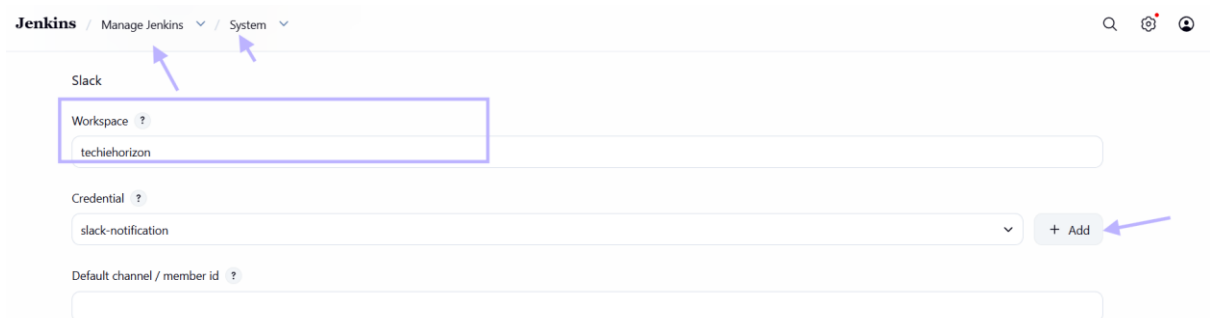
Jenkins CI is a customizable continuous integration server with over 600 plugins, allowing you to configure it to meet your needs.  
This integration will post build notifications to a channel in Slack.

**Post to Channel**  
Start by choosing a channel where Jenkins notifications will be posted.

devops-cloud-status-16

**Add Jenkins CI integration**

Go to Jenkins manage Jenkins > system configuration > select slack write the name of the organisation then click add credentials



Jenkins / Manage Jenkins / System

Slack

Workspace ?  
techiehorizon

Credential ?  
slack-notification

+ Add

Default channel / member id ?

And select secret text



Jenkins Credentials Provider: Jenkins

Domain  
Global credentials (unrestricted)

Kind  
Username with password  
Username with password  
GitHub App  
SSH Username with private key  
Secret file  
**Secret text**  
Certificate

Blank username; did you mean to use secret text credentials instead?  
☐ Treat username as secret

Password

Select token from slacks on step 3

### Step 3

After it's installed, click on **Manage Jenkins** again in the left navigation, and then go to **Configure System**. Find the **Global Slack Notifier Settings** section and add the following values:

- **Team Subdomain:** techiehorizon
- **Integration Token Credential ID:** Create a secret text credential using yAYWa4TWQKFfwP7ppazQMBwr as the value

The other fields are optional. You can click on the question mark icons next to them for more

Paste the token and provide ID and description

Scope ?

Global (Jenkins, nodes, items, all child items, etc) ▼

Secret

.....

ID ?

slack-notification2

Description ?

slack-notification2

Cancel Add

After adding test connection then save

Credential ?

slack-notification2 ▼ + Add

Default channel / member id ?

☐ Custom slack app bot user ?

Advanced ▼

Test Connection

Clear cache

Save Apply

Tested successfully



Go to manage Jenkins in GUI and then click system then setup slack details

Slack

Workspace ?

techiehorizon

Credential ?

slack-notification2 ▼ + Add

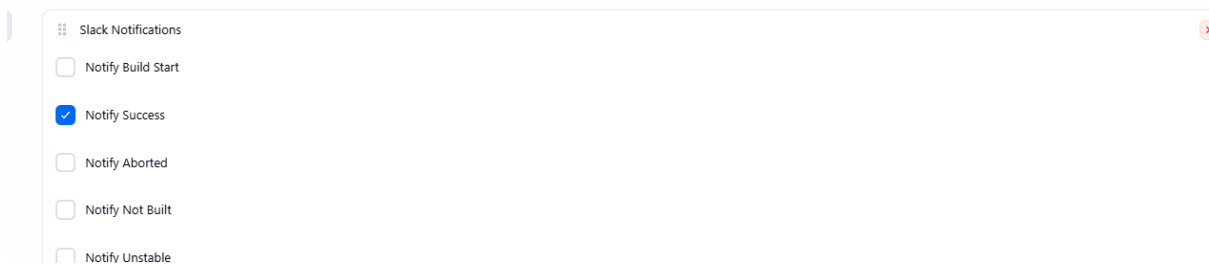
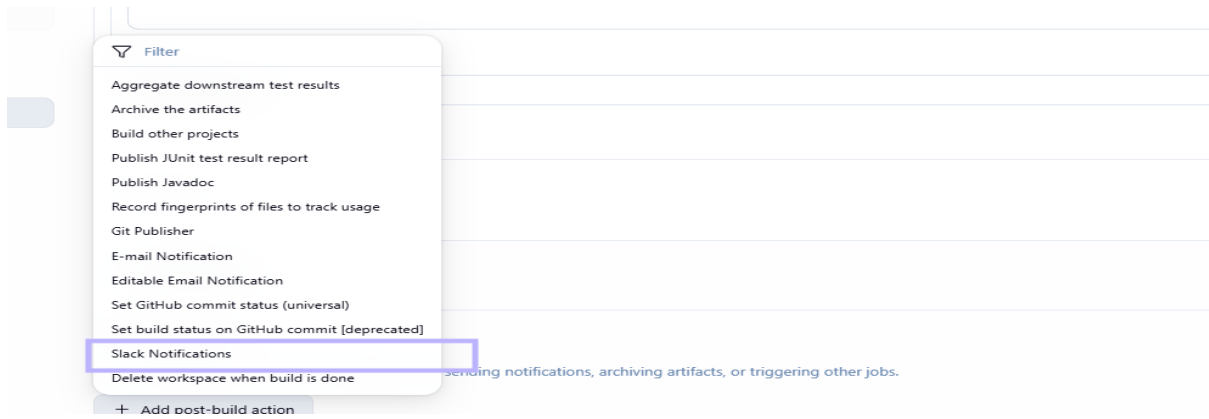
Default channel / member id ?

☐ Custom slack app bot user ?

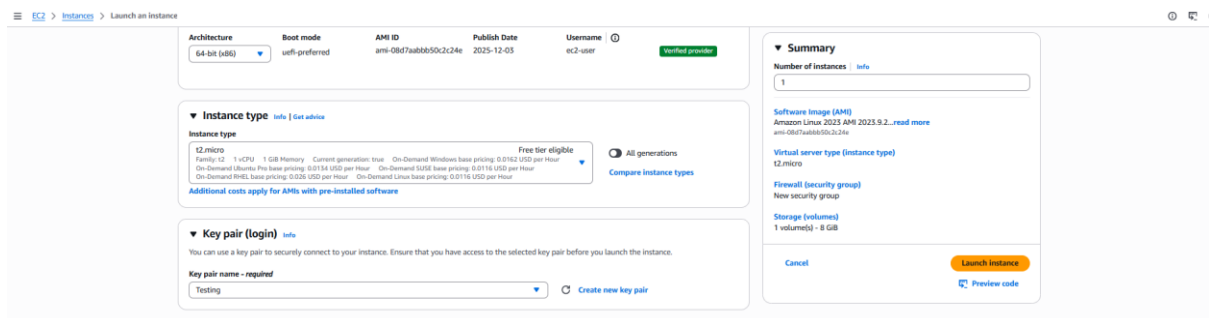
Advanced ▼

Test Connection

Go to job configure and select post build and select slac and choose type of notifications



## 6. Tomcat deployment



### Install Tomcat on EC2

- Install java 'sudo yum update -y'
- sudo yum install java-11-amazon-corretto-headless -y
- Run these commands on your EC2 server:
- cd /opt
- sudo wget https://dlcdn.apache.org/tomcat/tomcat-9/v9.0.87/bin/apache-tomcat-9.0.87.tar.gz
- sudo tar -xvf apache-tomcat-9.0.87.tar.gz
- sudo mv apache-tomcat-9.0.87 tomcat
- sudo chmod -R 755 /opt/tomcat

```

[ec2-user@ip-172-31-24-156 ~]$ sudo su -
[root@ip-172-31-24-156 ~]# cd /opt
[root@ip-172-31-24-156 opt]# wget https://dlcdn.apache.org/tomcat/tomcat-9/t9.0.113/bin/apache-tomcat-9.0.113.tar.gz
--2025-12-31 20:43:43-- https://dlcdn.apache.org/tomcat/tomcat-9/t9.0.113/bin/apache-tomcat-9.0.113.tar.gz
Resolving dlcdn.apache.org (dlcdn.apache.org)... 151.101.2.132, 2601:441:1444
Connecting to dlcdn.apache.org (dlcdn.apache.org)|151.101.2.132|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 13049663 (12M) [application/x-gzip]
Saving to: 'apache-tomcat-9.0.113.tar.gz'

apache-tomcat-9.0.113.tar.gz 100%[=====] 12.44M --.-20/s in 0.1s

2025-12-31 20:43:44 (103 MB/s) - 'apache-tomcat-9.0.113.tar.gz' saved [13049663/13049663]

[root@ip-172-31-24-156 opt]# ls
apache-tomcat-9.0.113.tar.gz
[root@ip-172-31-24-156 opt]# tar -xvf apache-tomcat-9.0.113.tar.gz
apache-tomcat-9.0.113/conf/
apache-tomcat-9.0.113/conf/catalina.policy
apache-tomcat-9.0.113/conf/catalina.properties
apache-tomcat-9.0.113/conf/context.xml
apache-tomcat-9.0.113/conf/jaspic-providers.xml
apache-tomcat-9.0.113/conf/jaspic-providers.xml
apache-tomcat-9.0.113/conf/logging.properties
apache-tomcat-9.0.113/conf/server.xml
apache-tomcat-9.0.113/conf/tomcat-users.xml
apache-tomcat-9.0.113/webapps/

```

## JAVA installation and Set java home

### Step 1 — Install Java 11 (Recommended for Tomcat 9)

- Run these commands:
- `sudo yum update -y`
- `sudo yum install java-11-amazon-corretto-headless -y`
- Verify:
- `java -version`
- Expected output:
- `openjdk version "11.x"`

### ✓ Step 2 — Set JAVA\_HOME

Find Java installation path:

- `sudo alternatives --display java`

Look for something like:

- `/usr/lib/jvm/java-11-amazon-corretto.x86_64`
- Then set JAVA\_HOME:
- `echo "export JAVA_HOME=/usr/lib/jvm/java-11-amazon-corretto.x86_64" | sudo tee -a /etc/profile`
- `echo "export PATH=$JAVA_HOME/bin:$PATH" | sudo tee -a /etc/profile`
- `source /etc/profile`
- Verify:
- `echo $JAVA_HOME`

### ✓ Step 3 — Restart Tomcat

- `sudo sh /opt/tomcat/bin/shutdown.sh`
- `sudo sh /opt/tomcat/bin/startup.sh`

```

NOTE: Picked up _JAVA_OPTIONS: --add-opens=java.base/java.lang=ALL-UNNAMED --add-opens=java.base/java.lang.invoke=ALL-UNNAMED --add-opens=java.base/java.lang.reflect=ALL-UNNAMED --add-opens=java.base/java.io=ALL-UNNAMED --add-opens=java.base/java.
Dec 31, 2025 9:07:09 PM org.apache.catalina.startup.Catalina stopServer
SEVERE: Could not contact [localhost:8005] (base port [8005] and offset [0]). Tomcat may not be running.
Dec 31, 2025 9:07:09 PM org.apache.catalina.startup.Catalina stopServer
SEVERE: Error stopping Catalina
java.net.ConnectException: Connection refused (Connection refused)
 at java.base/java.net.PlainSocketImpl.socketConnect(Native Method)
 at java.base/java.net.AbstractPlainSocketImpl.doConnect(AbstractPlainSocketImpl.java:412)
 at java.base/java.net.AbstractPlainSocketImpl.connectToAddress(AbstractPlainSocketImpl.java:255)
 at java.base/java.net.AbstractPlainSocketImpl.connect(AbstractPlainSocketImpl.java:237)
 at java.base/java.net.Socket.connect(Socket.java:593)
 at java.base/java.net.Socket.connect(Socket.java:609)
 at java.base/java.net.Socket.connect(Socket.java:568)
 at java.base/java.net.Socket.<init>(Socket.java:454)
 at java.base/java.net.Socket.<init>(Socket.java:231)
 at org.apache.catalina.startup.Catalina.stopServer(Catalina.java:630)
 at java.base/jdk.internal.reflect.NativeMethodAccessorImpl.invoke(Native Method)
 at java.base/jdk.internal.reflect.NativeMethodAccessorImpl.invoke(NativeMethodAccessorImpl.java:62)
 at java.base/jdk.internal.reflect.DelegatingMethodAccessorImpl.invoke(DelegatingMethodAccessorImpl.java:43)
 at java.base/java.lang.reflect.Method.invoke(Method.java:566)
 at org.apache.catalina.startup.Bootstrap.main(Bootstrap.java:350)
 at org.apache.catalina.startup.Bootstrap.main(Bootstrap.java:479)
Using CATALINA_BASE: /opt/tomcat
Using CATALINA_HOME: /opt/tomcat
Using CATALINA_TMPDIR: /opt/tomcat/temp
Using JRE_HOME: /usr
Using CLASSPATH: /opt/tomcat/bin/bootstrap.jar:/opt/tomcat/bin/tomcat-juli.jar
Tomcat started.

```

## Allow Tomcat Manager Login

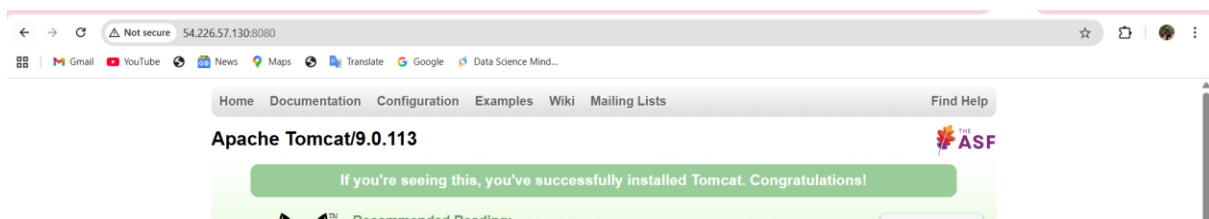
- Edit the user file:
- `sudo vi /opt/tomcat/conf/tomcat-users.xml`
- Add inside `<tomcat-users>` tag:
- `<role rolename="manager-script"/>`
- `<role rolename="manager-gui"/>`
- `<user username="deploy" password="deploy123" roles="manager-gui,manager-script"/>`

```

<?xml version="1.0" encoding="UTF-8"?>
<!--
Licensed to the Apache Software Foundation (ASF) under one or more
contributor license agreements. See the NOTICE file distributed with
this work for additional information regarding copyright ownership.
The ASF licenses this file to You under the Apache License, Version 2.0
(the "License"); you may not use this file except in compliance with
the License. You may obtain a copy of the License at
 http://www.apache.org/licenses/LICENSE-2.0
Unless required by applicable law or agreed to in writing, software
distributed under the License is distributed on an "AS IS" BASIS,
WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
See the License for the specific language governing permissions and
limitations under the License.
-->
<tomcat-users>
 <role rolename="manager-gui"/>
 <role rolename="manager-script"/>
 <user username="deploy" password="deploy123" roles="manager-gui,manager-script"/>
</tomcat-users>

```

## Tomcat is deployed



## Added user

Edit this file in this path `/opt/tomcat/conf/context.xml`

```

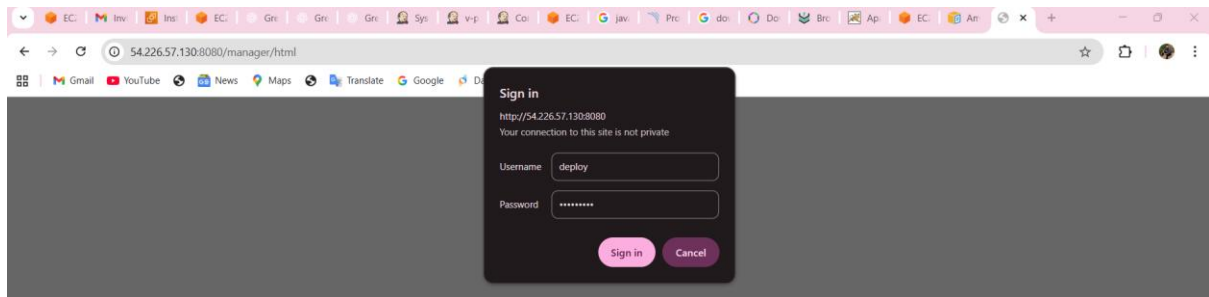
<?xml version="1.0" encoding="UTF-8"?>
<!--
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contributor license agreements. See the NOTICE file distributed with
this work for additional information regarding copyright ownership.
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distributed under the License is distributed on an "AS IS" BASIS,
WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
See the License for the specific language governing permissions and
limitations under the License.
-->
<Context antiResourceLocking="false" privileged="true" >
 <CookieProcessor className="org.apache.tomcat.util.http.Rfc6265CookieProcessor"
 sameSiteCookies="strict"/>
 <!-- <Valve className="org.apache.catalina.valves.RemoteIpValve"
 allow="127.0.0.0/8" ipHeader="x-forwarded-for"/> -->
 <!-- <Manager sessionAttributeValueClassNameFilter="java.lang.(?:Boolean|Integer|Long|Number|String)|org.apache.catalina.filters.CsvPreventionFilter$BaseCache(?:$|?)java.util.(?:LinkedHashMap|>
</Context>

```

Edit file in this path /opt/tomcat/conf/Catalina/localhost/<app-name>.xml

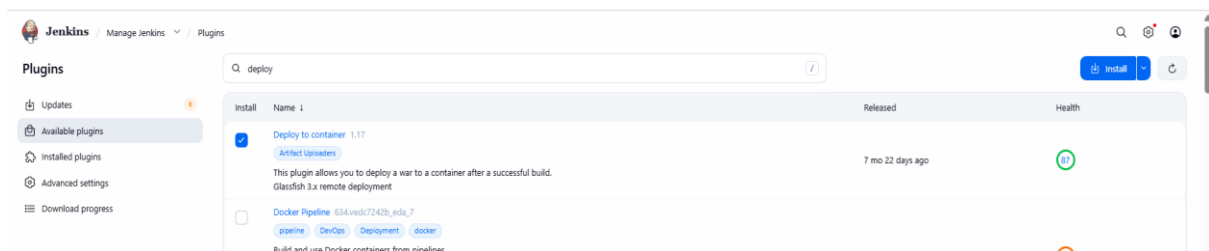
```
<?xml version="1.0" encoding="UTF-8"?>
<!--
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 contributor license agreements. See the NOTICE file distributed with
 this work for additional information regarding copyright ownership.
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 (the "License"); you may not use this file except in compliance with
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 distributed under the License is distributed on an "AS IS" BASIS,
 WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
 See the License for the specific language governing permissions and
 limitations under the License.
-->
<Context antiResourceLocking="false" privileged="true" >
 <CookieProcessor className="org.apache.tomcat.util.http.Rfc6265CookieProcessor"
 sameSiteCookies="strict" />
 <!-- <Valve className="org.apache.catalina.valves.RemoteAddrValve"
 allow="127.0.0.0/8:::/128" /> -->
 <!-- <Mount appMountAttributeValueType="java.lang.(?:Boolean|Integer|Long|Number|String)|org.apache.catalina.filters.CorsPreventionFilter|javax.cache.(?:[S|I|T|J]java).util.(?:Linked|HashMap|> -->
</Context>
```

Login under tomcat manager, as we setup already



Message:		OK			
Manager					
<a href="#">List Applications</a>		<a href="#">HTML Manager Help</a>		<a href="#">Manager Help</a>	
<a href="#">Server Status</a>					
Applications					
Path	Version	Display Name	Running	Sessions	Commands
/	None specified	Welcome to Tomcat	true	0	Start Stop Reload Undeploy   Expire sessions with idle ≥ 30 minutes
/docs	None specified	Tomcat Documentation	true	0	Start Stop Reload Undeploy   Expire sessions with idle ≥ 30 minutes
/examples	None specified	Servlet and JSP Examples	true	0	Start Stop Reload Undeploy

Plugin tomcat ‘Deploy to container’



Add credentials of tomcat

Manage Jenkins / Credentials / System / Global credentials (unrestricted)

### New credentials

Kind: Username with password

Scope: Global (Jenkins, nodes, items, all child items, etc)

Username: deploy

☒ Treat username as secret

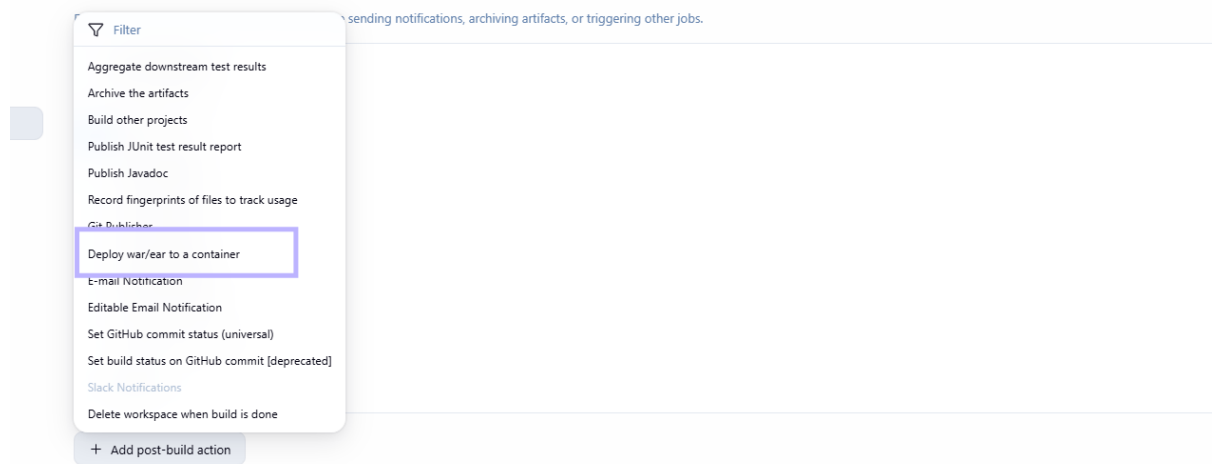
Password: \*\*\*\*\*

ID: tomcat-deploy

Description: tomcat-deploy

Create

Go to job and select post build option then select Deploy war



Following details added in the job configure under Deploy war

Deploy war/ear to a container

WAR/EAR files: \*\*/\*.war

Context path: /hiring

Containers

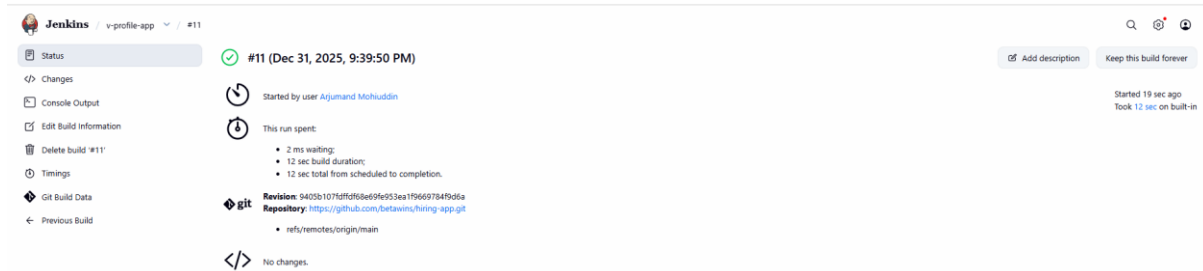
Tomcat 9.x Remote

Credentials: deploy/\*\*\*\*\* (tomcat-deploy)

Tomcat URL: http://54.226.57.130:8080/manager/text

Advanced

+ Add Container

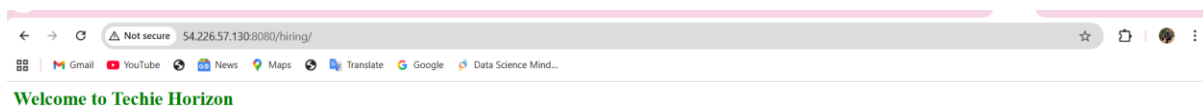


**web application is now:**

Successfully built in Jenkins

Deployed to Tomcat

Accessible via browser



### **Task takeaway:**

- ✓ GitHub →
- ✓ Jenkins CI Build →
- ✓ Nexus Repository Upload →
- ✓ Tomcat Deployment →
- ✓ Web App Accessible to the World

Developer Push Code → GitHub → Jenkins Webhook Trigger

→ Maven Build in Jenkins

→ SonarQube Quality Analysis

→ Nexus Artifact Upload (.war)

→ Slack notification on every successful job execution

→ Jenkins Deployment Job

→ Tomcat Server Running Latest App

**2. Setup a Jenkins CICD pipeline using Declarative pipeline using feature-1.1 branch. [https://github.com/betawins/sabear\\_simplecutomerapp/tree/feature-1.1](https://github.com/betawins/sabear_simplecutomerapp/tree/feature-1.1)**

**stages:-**

**Git Clone**

**SonarQube Integration**



**Maven Compilation**

**Nexus Artifactory**

**Slack Notification**

**Deploy On tomcat**

## CI/CD Pipeline Setup – Steps Followed & Troubleshooting Report

### ◆ Goal

Automate build → code quality check → artifact upload → deploy to Tomcat using:

- **Jenkins**
  - **GitHub**
  - **SonarQube**
  - **Nexus**
  - **Tomcat Server**
- 

### 1 Jenkins – Pipeline Setup Steps

#### ✓ Create Pipeline

- Installed required plugins:
  - Pipeline, Git, SonarQube Scanner, Nexus Artifact Uploader, SSH Agent, Slack Notifications.
- Added global tools configuration:
  - **JDK** (Java 17)
  - **Maven** (3.x)

#### ✓ Configure Credentials

Service	Type	ID Used	Purpose
GitHub	None (public repo)	—	Code pull
Nexus	Username + Password	nexus-cred	WAR upload to Nexus

Service	Type	ID Used	Purpose
Tomcat SSH	SSH Private Key	tomcat-ssh	Copy WAR + restart Tomcat
Slack	Token Credential	slack-notification2	Notification

### ✓ SonarQube Integration

- Added server details in **Manage Jenkins → Credentials → Global**
- Added global setting:
  - Name: *SonarQube*
  - Token + URL:
  - `http://18.206.205.127:9000`

### ✓ Nexus Repository Setup

- Created repository:
  - Type: **maven2 → snapshots**
  - URL used in Jenkins:
  - `http://44.210.91.55:8081/repository/maven-snapshots/`

### ✓ Provided Maven Settings in Jenkins

- Added **Global settings.xml** in *Managed Files*
- It included repository credentials securely.

## 2 Jenkins Pipeline Script

Pipeline stages successfully configured:

- 1 Git Clone
- 2 SonarQube Analysis
- 3 Maven Build
- 4 Deploy to Nexus
- 5 Deploy to Tomcat
- 6 Slack Notification

Final Deploy script fix:

```
scp -o StrictHostKeyChecking=no -i $SSH_KEY target/*.war ec2-user@13.222.160.53:/opt/tomcat/webapps/
```

```
ssh -o StrictHostKeyChecking=no -i $SSH_KEY ec2-user@13.222.160.53 "sudo /opt/tomcat/bin/shutdown.sh || true; sudo /opt/tomcat/bin/startup.sh"
```

---

### Troubleshooting Steps Performed

Issue	Cause	Fix Applied
401 Unauthorized while uploading to Nexus	Wrong credentials in settings.xml	Correct password updated
"No credentials ID found: nexus-creds"	Wrong ID in pipeline	Updated to correct ID: nexus-cred
Permission denied on Tomcat WAR upload	Insufficient permissions on /opt/tomcat/webapps	sudo chmod -R 777 /opt/tomcat/webapps
systemctl restart tomcat failed	Tomcat manually installed, not a service	Used shutdown.sh/startup.sh instead
Slack notification failed	URL missing	Configured Slack Token + Channel
SonarQube token issue initially	Scanner not configured in Jenkins	Added SonarQube environment in pipeline

---

### Final Status ✓

Stage	Result
Git Clone	Success
SonarQube Code Analysis	Success ✓
Maven Build	Success ✓
Nexus Upload	Success ✓
Tomcat Deployment	Success ✓
Slack Notification	Success ✓

---

### Outcome

Fully automated CI/CD pipeline from **GitHub** → **Jenkins** → **SonarQube** → **Nexus** → **Tomcat**

Application successfully deployed after each pipeline run 🚀🔥

```
[ec2-user@ip-172-31-21-117 ~]$ systemctl start jenkins.service
Failed to start jenkins.service: Access denied
See system logs and 'systemctl status jenkins.service' for details.
[ec2-user@ip-172-31-21-117 ~]$ sudo systemctl start jenkins.service
[ec2-user@ip-172-31-21-117 ~]$ sudo systemctl status jenkins
jenkins.service - Jenkins Continuous Integration Server
Loaded: loaded (/usr/lib/systemd/system/jenkins.service; disabled; preset: disabled)
Drop-In: /etc/systemd/system/jenkins.service.d
└─/override.conf
Active: active (running) since Thu 2026-01-01 11:35:58 UTC; 16s ago
Main PID: 2442 (java)
Tasks: 56 (limit: 2205)
Memory: 476.9M
CGroup: /system.slice/jenkins.service
└─/usr/bin/java -Djava.io.tmpdir=/var/lib/jenkins/tmp -jar /usr/share/java/jenkins.war --webroot=/var/cache/jenkins/war --httpPort=8081

Jan 01 11:35:58 ip-172-31-21-117.ec2.internal jenkins[2442]: 2026-01-01 11:35:58.977+0000 [id=30] INFO h.p.b.g.GlobalTimeOutConfigurationLoad: global timeout not set
Jan 01 11:35:58 ip-172-31-21-117.ec2.internal jenkins[2442]: 2026-01-01 11:35:58.978+0000 [id=31] INFO hudson.slaves.SlaveComputer$tryReconnect: Attempting to reconnect Agent-1
Jan 01 11:35:58 ip-172-31-21-117.ec2.internal jenkins[2442]: 2026-01-01 11:35:58.978+0000 [id=31] INFO hudson.slaves.SlaveComputer$tryReconnect: Attempting to reconnect slave-2
Jan 01 11:35:58 ip-172-31-21-117.ec2.internal jenkins[2442]: 2026-01-01 11:35:58.979+0000 [id=30] INFO jenkins.InitReactorRunner$1onAttained: System config loaded
Jan 01 11:35:58 ip-172-31-21-117.ec2.internal jenkins[2442]: 2026-01-01 11:35:58.979+0000 [id=30] INFO jenkins.InitReactorRunner$1onAttained: System config adapted
Jan 01 11:35:58 ip-172-31-21-117.ec2.internal jenkins[2442]: 2026-01-01 11:35:58.980+0000 [id=30] INFO jenkins.InitReactorRunner$1onAttained: Loaded all jobs
Jan 01 11:35:58 ip-172-31-21-117.ec2.internal jenkins[2442]: 2026-01-01 11:35:58.980+0000 [id=31] INFO jenkins.InitReactorRunner$1onAttained: Configuration for all jobs updated
Jan 01 11:35:58 ip-172-31-21-117.ec2.internal jenkins[2442]: 2026-01-01 11:35:58.980+0000 [id=31] INFO jenkins.InitReactorRunner$1onAttained: Completed initialization
Jan 01 11:35:58 ip-172-31-21-117.ec2.internal jenkins[2442]: 2026-01-01 11:35:58.980+0000 [id=23] INFO hudson.lifecycle.Lifecycle$onReady: Jenkins is fully up and running
Jan 01 11:35:58 ip-172-31-21-117.ec2.internal systemd[1]: Started jenkins.service - Jenkins Continuous Integration Server.
[ec2-user@ip-172-31-21-117 ~]$ cd /opt
[ec2-user@ip-172-31-21-117 ~]$ ls
sonar-scanner-4.4.2-Linux.x86_64.rpm sonar-scanner
```

```
[root@ip-172-31-92-107 bin]# cd ..
[root@ip-172-31-92-107 /]# cd ..
[root@ip-172-31-92-107 /]# ls
bin boot dev etc home lib lib64 local media mnt opt proc root run sbin srv sys usr var
[root@ip-172-31-92-107 /]# cd ..
[root@ip-172-31-92-107 /]# cd .. / ..
bin boot dev etc home lib lib64 local media mnt opt proc root run sbin srv sys usr var
[root@ip-172-31-92-107 /]# ls
bin boot dev etc home lib lib64 local media mnt opt proc root run sbin srv sys usr var
[root@ip-172-31-92-107 /]# cd ..
[root@ip-172-31-92-107 /]# docker start sonarqube
sonarqube
[root@ip-172-31-92-107 /]# docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS
d8b949c8f1e sonarqube:5.9.0 "/opt/sonarqube/bock-" 24 hours ago Up 12 seconds 0.0.0.0:9000->9000/tcp, :::9000->9000/tcp
[root@ip-172-31-92-107 /]# ||
```

```
i-01776552a99740f63 (sonarqube_with_docker)

last login: Wed Dec 31 18:20:14 UTC 2025 on pta/1
[root@ip-172-31-84-50 ~]$ systemctl restart nexus
[root@ip-172-31-84-50 ~]$ systemctl status nexus -l
nexus.service - nexus service
Loaded: loaded (/usr/lib/systemd/system/nexus.service; enabled; preset: disabled)
Active: active (running) since Thu 2026-01-01 11:49:53 UTC; 46s ago
Process: 4315 ExecStart=/opt/nexus/bin/nexus start (code=exited, status=0/SUCCESS)
Main PID: 4499 (java)
Tasks: 63 (limit: 9432)
Memory: 1.9G
CGroup: /system.slice/nexus.service
└─4499 /usr/lib/jvm/java/bin/java -server -Dinstall4j.jvmDir=/usr/lib/jvm/java -Dext4j.moduleName=/opt/nexus/bin/nexus -XX:+UnlockDiagnosticVMOptions -Dinstall4j.launcherId=245 -Dinstall4j.swc=false -Dl4jv=0 -Dl4jv=0 -Dl4jv=0 -Dl4jv=0

Jan 01 11:49:53 ip-172-31-84-50.ec2.internal systemd[1]: Starting nexus.service - nexus service...
Jan 01 11:49:53 ip-172-31-84-50.ec2.internal nexus[4315]: Starting nexus
Jan 01 11:49:53 ip-172-31-84-50.ec2.internal systemd[1]: Started nexus.service - nexus service.
lines 1-14/14 (END)
```

```
i-045f79db1c7bd5347 (nexus)

PublicIP: 44.210.91.55 PrivateIP: 172.31.84.50

last login: Wed Dec 31 20:42:41 2025 from 18.206.107.27
[ec2-user@ip-172-31-24-156 ~]$ ls
last login: Wed Dec 31 20:43:22 UTC 2025 on pta/1
[ec2-user@ip-172-31-24-156 ~]$ cd /opt/tomcat/bin/
[root@ip-172-31-24-156 bin]# ls
bootstrap.jar catalina.bat ciphera.bat common-logging-native.tar.gz configtest.bat daemon.sh digest.sh mahadash.sh setclasspath.sh shutdown.sh startup.sh tcat-native.tar.gz tool-wrapper.sh version.sh
catalina-logging.xml catalina.sh ciphera.sh common-logging.jar configtest.sh digest.sh mahadash.bat setclasspath.bat shutdown.bat startup.bat tcat-juli.jar tool-wrapper.bat version.bat
[root@ip-172-31-24-156 bin]# ./startup.sh h
Using CATALINA_HOME: /opt/tomcat
Using CATALINA_BASE: /opt/tomcat
Using CATALINA_TMPDIR: /opt/tomcat/temp
Using JRE_HOME: /usr/lib/jvm/java-11-amazon-corretto.x86_64
Using CLASSPATH: /opt/tomcat/bin/bootstrap.jar:/opt/tomcat/bin/tomcat-juli.jar
tomcat started.
[root@ip-172-31-24-156 bin]# ./startup.sh
Using CATALINA_HOME: /opt/tomcat
Using CATALINA_BASE: /opt/tomcat
Using CATALINA_TMPDIR: /opt/tomcat/temp
Using JRE_HOME: /usr/lib/jvm/java-11-amazon-corretto.x86_64
Using CLASSPATH: /opt/tomcat/bin/bootstrap.jar:/opt/tomcat/bin/tomcat-juli.jar
tomcat started.
[root@ip-172-31-24-156 bin]# ||
```

Default is http://localhost:9000

Server authentication token

SonarQube authentication token. Mandatory when anonymous access is disabled.

jenkins-pipeline

▼

+ Add

Advanced ▼

Save

Apply

```

#Configure here general information about the environment, such as SonarQube server connection details for example
#No information about specific project should appear here

#----- Default SonarQube server
sonar.host.url=http://19.204.205.137:9000

#----- Default source code encoding
sonar.sourceEncoding=UTF-8

```

"We configure NEXUS URL and credentials in ~/.m2/settings.xml so Maven on Jenkins knows where to deploy artifacts securely. It prevents exposing credentials in source code and allows CI/CD to upload WAR files automatically."

```

[root@ip-172-31-21-117 .m2]# cd repository/
[root@ip-172-31-21-117 repository]# ls
backport-util-concurrent classworlds.xml commons-cli commons-io commons-logging junit log4j org
[root@ip-172-31-21-117 repository]# cd ..
[root@ip-172-31-21-117 .m2]# ls
repository
[root@ip-172-31-21-117 .m2]# cd /var/lib/jenkins/.m2
sudo touch settings.xml
[root@ip-172-31-21-117 .m2]# vi settings.xml ||

```

i-0f506b3638890d1e4 (Master\_jenkins)  
DuJdZrDe 4x85 11x 137 DiskUsedDe 172 31 21 117

Added script and credentials

```

settings.xml<?xml="http://maven.apache.org/SETTINGS/1.0.0"
xmlns:xi="http://www.w3.org/2001/XMLSchema-instance"
xi:schemaLocation="http://maven.apache.org/SETTINGS/1.0.0
http://maven.apache.org/xsd/settings-1.0.0.xsd">

<servers>
<server>
<id>sonar-cred</id>
<username>admin</username>
<password>8C0200b8123</password>
</server>
</servers>

<mirrors>
<mirror>
<id>sonar</id>
<mirrorOf>~</mirrorOf>
<url>http://44.210.91.55:8081/repository/maven-releases</url>
</mirror>
</mirrors>
</settings>

```

Script is given below:

pipeline {

agent any

environment {

SONARQUBE = 'SonarQube'

GIT\_REPO = "https://github.com/betawins/sabear\_simplecutomerapp.git"

TOMCAT\_SERVER = "ec2-user@13.222.160.53"

}

stages {

stage('Git Clone') {

```

steps {
 git branch: 'feature-1.1',
 url: "${GIT_REPO}"
}
}

stage('SonarQube Integration') {
 steps {
 withSonarQubeEnv("${SONARQUBE}") {
 sh "mvn clean verify sonar:sonar -DskipTests"
 }
 }
}

stage('Maven Compilation') {
 steps {
 sh "mvn clean install -DskipTests"
 }
}

stage('Nexus Artifactory') {
 steps {
 withCredentials([usernamePassword(credentialsId: 'Nexus_server',
 usernameVariable: 'NEXUS_USER',
 passwordVariable: 'NEXUS_PASS')]) {
 sh ""
 mvn deploy -DskipTests \

```

```
-
DaltDeploymentRepository=nexus::default::http://44.210.91.55:8081/repository/maven-
snapshots/\
```

```
-Drepo.username=$NEXUS_USER -Drepo.password=$NEXUS_PASS
```

```
""
```

```
}
```

```
}
```

```
}
```

```
stage('Deploy On Tomcat') {
```

```
 steps {
```

```
 withCredentials([sshUserPrivateKey(credentialsId: 'tomcat-ssh',
 keyFileVariable: 'SSH_KEY')]) {
```

```
 sh ""
```

```
 scp -o StrictHostKeyChecking=no -i $SSH_KEY target/*.war ec2-
user@13.222.160.53:/opt/tomcat/webapps/
```

```
 ssh -o StrictHostKeyChecking=no -i ${SSH_KEY} ec2-user@13.222.160.53 "sudo
/opt/tomcat/bin/shutdown.sh || true; sudo /opt/tomcat/bin/startup.sh"
```

```
""
```

```
}
```

```
}
```

```
}
```

```
stage('Slack Notification') {
```

```
 steps {
```

```
 slackSend(channel: '#cicd-alerts',
```

```
 message: "Deployment Success 🟢 — ${env.JOB_NAME}
#${env.BUILD_NUMBER}",
```

```

 color: 'good')
 }
}
}

post {
 failure {
 slackSend(channel: '#cicd-alerts',
 message: "Deployment Failed ❌ — ${env.JOB_NAME}
#${env.BUILD_NUMBER}",
 color: 'danger')
 }
}
}

```

#### Definition

Pipeline script

Script ?

```

37
38 stage('Deploy On Tomcat') {
39 steps {
40 withCredentials([sshUserPrivateKey(credentialsId: 'tomcat-ssh', keyFileVariable: 'SSH_KEY')]) {
41 sh '''
42 scp -o StrictHostKeyChecking=no -i $SSH_KEY target/*.war ec2-user@13.222.160.53:/opt/tomcat/webapp
43 ssh -o StrictHostKeyChecking=no -i $SSH_KEY ec2-user@13.222.160.53 "sudo systemctl restart tomcat
44 '''
45 }
46 }
47 }
48
49 stage('Slack Notification') {
50 steps {
51

```

☒ Use Groovy Sandbox ?

Save

Apply



Global (Jenkins, nodes, items, all child items, etc)

ID ?

tomcat-ssh

Description ?

tomcat-ssh

Username

ec2-user

☐ Treat username as secret ?

Private Key

Enter directly

Key

Enter New Secret Below

-----BEGIN RSA PRIVATE KEY-----  
MIIIEowIBAAKCAQEA31DfN7Hfdr1j73ufedv1y+5YA4SYQ23GvYnPg8h8ac2ZB  
19g5jDWeq1261gop7378Q1Z6F3n+LatMQga9HxtjLu4tFnc8Bv1XtyjX+2328  
QP25RkF37W6p8v6/SSNMdaJVTYL3211cUovd1gT0hyY76uE2w7U8ap7EMvUlp9  
vN1R8Z1NPNfZ3yJk9Rv9sR3ja98coVePHPTAL3X0oxBPwh1ycno0sInuL/p8m7  
dyMnT84JohfahjFvkt8noFh0vvQQVBAdKhJ595x3xG5KX21mDyngR/Ic+wn8ykSt  
35F7UmHqjqu8+b91epo8pn20bYlx5wvT8cxQIDAQ46A01BA6zK1g/q8m1308Kj  
+8gcn5hc7VMg3jD2EzChp6/nhouT8wKwV8pD0521OCVP30qplm60FUjC/Nz  
0wv71FBwvrs5n+mh1/qa3h8/NS3XC1QJ555bJ/wgcA5VA9FJ0dK8C31s7Dq/F5bv4  
n1NneAM7eSH4K5dN717XCR101nkn1vTovrdRvHk+arERT1nCRGIdmlutA/nk+h

Create

```
<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0"
 xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
 xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/maven-v4_0_0.xsd">
 <modelVersion>4.0.0</modelVersion>
 <groupId>com.jaredpalmer</groupId>
 <artifactId>SimpleCustomerApp</artifactId>
 <packaging>war</packaging>
 <version>4.0.0</version>
 <name>SimpleCustomerApp</name>
 <url>http://maven.apache.org/</url>
 <properties>
 <maven.compiler.source>1.8</maven.compiler.source>
 <maven.compiler.target>1.8</maven.compiler.target>
 </properties>
 <build>
 <plugins>
 <plugin>
 <groupId>org.sonarsource.scanner.maven</groupId>
 <artifactId>sonar-maven-plugin</artifactId>
 <version>3.3.1.2184</version>
 </plugin>
 </plugins>
 </build>
</project>
```

Jenkins

Manage Jenkins

Managed files

Config File Management

You can edit or remove your configuration files

Jenkins

Task2-CICD-Feature-1.1

#29

Status

Changes

Console Output

Edit Build Information

Delete build #29

Timings

Git Build Data

Pipeline Overview

Restart from Stage

Replay

Pipeline Steps

Workspaces

Previous Build

#29 (Jan 1, 2026, 6:57:38 PM)

Started by user Arjumand Mohiuddin

This run spent:

- 16 ms waiting;
- 27 sec build duration;
- 27 sec total from scheduled to completion;

Revision: 05222ef4b498f0dd73a8a1b6fbd0c619d398a

Repository: https://github.com/betawins/sabear\_simplecustomerapp.git

- refs/remotes/origin/feature-1.1

The following steps that have been detected may have insecure interpolation of sensitive variables (click here for an explanation):

- sh: [INDEX\_PASS]
- sh: [SSH\_KEY]

No changes.

Add description

Keep this build forever

Started 50 sec ago

Took 27 sec

←

→

↺

⚠ Not secure

13.222.160.53:8080/SimpleCustomerApp-29-SNAPSHOT/

☆

🖨

🔍

⋮

Gmail

YouTube

News

Maps

Translate

Google

Data Science Mind...

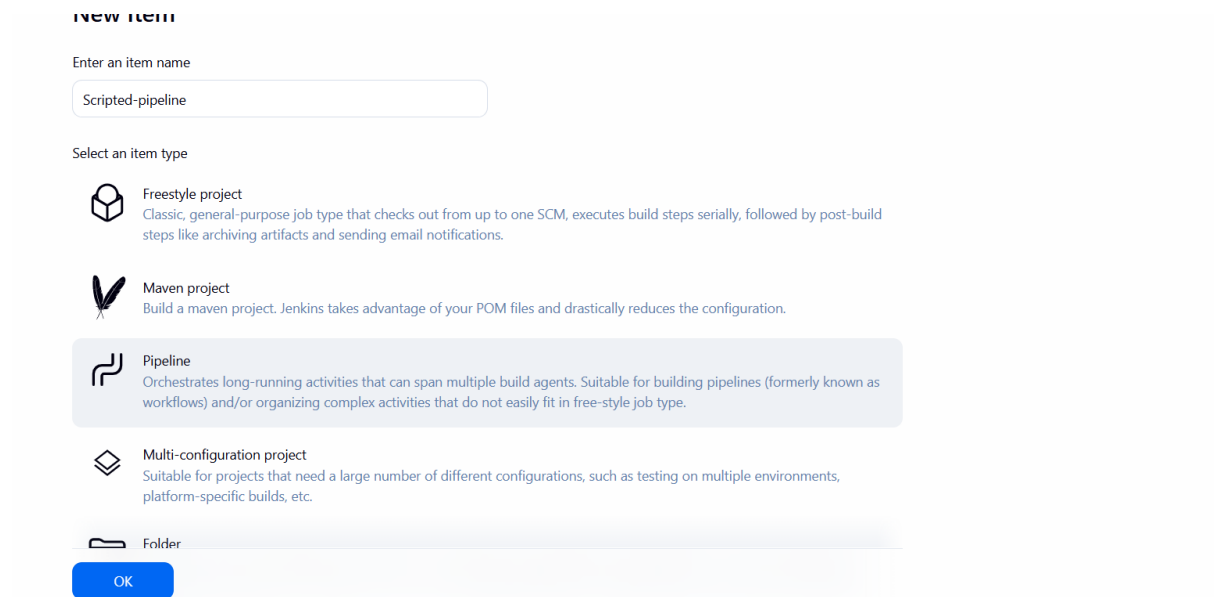
🎉 Deployment Successful 🎉

Welcome to Simple Customer App

### 3. Setup a Jenkins CI/CD pipeline using Scripted pipeline using feature-1.1 branch. [https://github.com/betawins/sabear\\_simplecutomerapp/tree/feature-1.1](https://github.com/betawins/sabear_simplecutomerapp/tree/feature-1.1)

#### stages:

1. Git Clone
2. SonarQube Integration
3. Maven Compilation
4. Nexus Artifactory
5. Slack Notification
6. Deploy On tomcat



**NEW ITEM**

Enter an item name

Scripted-pipeline

Select an item type

-  **Freestyle project**  
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.
-  **Maven project**  
Build a maven project. Jenkins takes advantage of your POM files and drastically reduces the configuration.
-  **Pipeline**  
Orchestrates long-running activities that can span multiple build agents. Suitable for building pipelines (formerly known as workflows) and/or organizing complex activities that do not easily fit in free-style job type.
-  **Multi-configuration project**  
Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.
-  **Folder**

OK

#### Scripted pipeline script:

```
node {

 def GIT_URL = "https://github.com/betawins/sabear_simplecutomerapp.git"
 def BRANCH = "feature-1.1"
 def NEXUS_URL = "http://34.239.127.252:8081/repository/maven-snapshots/"
 def TOMCAT_HOST = "98.93.21.135"
 def TOMCAT_PATH = "/opt/tomcat/webapps"

 stage('Git Clone') {
 git branch: BRANCH, url: GIT_URL
 }
}
```

```

stage('SonarQube Integration') {
 withSonarQubeEnv('SonarQube') {
 sh '''
 mvn clean verify sonar:sonar -DskipTests
 '''
 }
}

```

```

stage('Maven Compilation') {
 sh '''
 mvn clean install -DskipTests
 '''
}

```

```

stage('Nexus Artifactory') {
 withCredentials([usernamePassword(
 credentialsId: 'Nexus_server',
 usernameVariable: 'NEXUS_USER',
 passwordVariable: 'NEXUS_PASS'
)]) {
 sh '''
 mvn deploy -DskipTests \
 -
 DaltDeploymentRepository=nexus::default::http://34.239.127.252:8081/repository/
 maven-snapshots/
 '''
 }
}

```

The screenshot shows the Jenkins web interface for a pipeline named 'Scripted-pipeline'. On the left is a sidebar with navigation links: Status, Changes, Build Now, Configure, Delete Pipeline, SonarQube, Stages, Rename, and Pipeline Syntax. The main content area shows the pipeline's status as 'Scripted-pipeline' with a green checkmark. Below this, it displays 'SonarQube Quality Gate' results for 'SimpleCustomerApp', which is 'Passed'. It also shows 'server-side processing' as 'Success'. A 'Permalinks' section lists recent build events with timestamps. At the bottom left, there is a 'Builds' link with a dropdown arrow.

## Declarative Pipeline – Skeleton

```
pipeline {
 agent any

 environment {
 // Environment variables (optional)
 APP_NAME = "sample-app"
 }

 stages {

 stage('Git Clone') {
 steps {
 echo 'Cloning source code'
 }
 }

 stage('Build') {
 steps {
 echo 'Building application'
 }
 }

 stage('Test') {
 steps {
 echo 'Running tests'
 }
 }

 stage('Deploy') {
 steps {
 echo 'Deploying application'
 }
 }
 }

 post {
 success {
 echo 'Pipeline executed successfully'
 }
 failure {
 echo 'Pipeline failed'
 }
 }
}
```

---



## 2 Scripted Pipeline – Skeleton (IMPORTANT FOR YOUR TASK)

```
node {

 stage('Git Clone') {
 echo 'Cloning source code'
 }

 stage('Build') {
 echo 'Building application'
 }
}
```

```

stage('Test') {
 echo 'Running tests'
}

stage('Deploy') {
 echo 'Deploying application'
}
}

```

---

### ✓ 3 Scripted Pipeline Skeleton (with variables)

```

node {

 def APP_NAME = "sample-app"

 stage('Checkout') {
 echo "Checking out ${APP_NAME}"
 }

 stage('Compile') {
 echo "Compiling ${APP_NAME}"
 }

 stage('Package') {
 echo "Packaging ${APP_NAME}"
 }

 stage('Deploy') {
 echo "Deploying ${APP_NAME}"
 }
}

```

---

### ✓ 4 Minimal Skeleton (Very short – interview style)

```

node {
 stage('Stage-1') { }
 stage('Stage-2') { }
 stage('Stage-3') { }
}

```

**5. Create a parameterized job in Jenkins.** <https://github.com/betawins/spring3-mvc-maven-xml-hello-world-1.git>

## New Item

Enter an item name

Parameterized-Pipeline

Select an item type



Freestyle project

Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.



Maven project

Build a maven project. Jenkins takes advantage of your POM files and drastically reduces the configuration.



Pipeline

Orchestrates long-running activities that can span multiple build agents. Suitable for building pipelines (formerly known as workflows) and/or organizing complex activities that do not easily fit in free-style job type.



Multi-configuration project

Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.



Folder

OK

Configure

Pipeline script from SCM

SCM ?

Git

Repositories ?

Repository URL ?

https://github.com/betawins/spring3-mvc-maven-xmi-hello-world-1.git

Credentials ?

- none -

+ Add

Advanced ▾

+ Add Repository

Branches to build ?

Branch Specifier (blank for 'any') ?

\*/master

+ Add Branch

Repository browser ?

(Auto)

Additional Behaviours

+ Add

Script Path ?

Jenkinsfile\_parameterized

☒ Lightweight checkout ?

[Pipeline Syntax](#)

Advanced

Save

Apply

## Script:

```
node {
```

```
 see = "spring3-mvc-maven-xml-hello-world-1"
```

```
 stage('Git Clone') {
```

```
 git branch: params.BRANCH,
```

```
 credentialsId: 'github-creds',
```

```
 url: 'https://github.com/betawins/spring3-mvc-maven-xml-hello-world-1.git'
```

```
 }
```

```
 stage('Maven Build') {
```

```
 if (params.RUN_MAVEN_BUILD) {
```

```
 sh 'mvn clean package -DskipTests'
```

```
 } else {
```

```
 echo "Maven build skipped"
```

```
 }
```

```
 }
```

```
 stage('Deploy (Dummy)') {
```

```
 if (params.DEPLOY) {
```

```
 echo "Deploy stage enabled (demo purpose)"
```

```
 sh 'ls -l target | | true'
```

```
 } else {
```

```
 echo "Deploy skipped"
```

```
 }
```

```
 }
```

```
}
```

The screenshot displays the Jenkins web interface for a pipeline named 'Parameterized Pipeline'. The left sidebar contains navigation links: Status, Changes, Console Output, Edit Build Information, Delete build #3, Parameters, Timings, Git Build Data, Pipeline Overview, Replay, Pipeline Steps, Workspaces, and Previous Build. The main panel shows the execution details for build #3, dated Jan 2, 2026, at 11:22:06 AM. It indicates the build was started by user 'Arjmand Mohiuddin'. A summary of the run shows 0 ms waiting, 0.78 sec build duration, and 0.78 sec total from scheduled to completion. The 'git' step is highlighted, showing the repository URL 'https://github.com/betawins/spring3-mvc-maven-xml-hello-world-1.git' and the branch 'refs/remotes/origin/master'. The 'No changes' status is indicated. On the right, there are buttons for 'Add description' and 'Keep this build forever', along with a timestamp 'Started 9.4 sec ago' and 'Took 0.78 sec'.

## 5. Setup one slave machine for Jenkins.

### Install java 17

```
root@ip-172-31-24-55:~# sudo dnf install java-21-amazon-corretto-devel -y
Amazon Linux 2023 kernel livepatch repository
Dependencies resolved.
```

Package	Architecture	Version	Repository	Size
Installing:				
java-21-amazon-corretto-devel	x86_64	1:21.0.9+11-1.amzn2023.1	amazonlinux	149 k
Installing dependencies:				
alsa-lib	x86_64	1:2.7.2-1.amzn2023.0.2	amazonlinux	504 k
java-aws-sdk	noarch	1.12.10-1.amzn2023.0.2	amazonlinux	717 k
java-aws-sdk-bom	noarch	2.97.16.amzn2023.0.2	amazonlinux	1.3 M
java-aws-sdk-core	noarch	2.97.16.amzn2023.0.2	amazonlinux	447 k
java-aws-sdk-integration	noarch	2.97.16.amzn2023.0.2	amazonlinux	1.0 M
fontconfig	x86_64	2.13.9-4.amzn2023.0.2	amazonlinux	219 k
fontconfig-devel	noarch	1:2.0.5-12.amzn2023.0.2	amazonlinux	9.5 k
fontconfig-filesystem	noarch	2.13.9-4.amzn2023.0.2	amazonlinux	423 k
google-auto-fuse-common	noarch	20240401-1.amzn2023.0.2	amazonlinux	17 k
google-auto-fuse-vfs-fuse	noarch	20240401-1.amzn2023.0.2	amazonlinux	849 k
glibc	x86_64	1:2.37-1.amzn2023.0.2	amazonlinux	51 k

### Create ssh key and transfer it to authenticator\_keys

```
root@ip-172-31-24-55:~# ssh-keygen -t rsa -b 4096 -C 'root@ip-172-31-24-55'
Generating public/private rsa key pair.
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /root/.ssh/id_rsa.
Your public key has been saved in /root/.ssh/id_rsa.pub.
The key fingerprint is:
SHA256:172-31-24-55
The key's raw representation in hexadecimal is:
root@ip-172-31-24-55:~# cat /root/.ssh/id_rsa.pub
ssh-rsa AAAAB3NzaG1zb2EAAAADAQABAAQDAQ== root@ip-172-31-24-55
root@ip-172-31-24-55:~# scp /root/.ssh/id_rsa.pub i-0812067ae399318e6:/home/ubuntu/.ssh/authorized_keys
i-0812067ae399318e6 (Slave_instance)
```

### Go to master slave and and skey-scan

```
at java.base/jdk.internal.reflect.NativeMethodAccessorImpl.invoke(NativeMethodAccessorImpl.java:77)
at java.base/jdk.internal.reflect.DelegatingMethodAccessorImpl.invoke(DelegatingMethodAccessorImpl.java:43)
at java.base/java.lang.reflect.Method.invoke(Method.java:569)
at com.amazonaws.services.s3.internal.Main.main(Main.java:335)
[root@ip-172-31-24-55:~#]# ssh -i /root/.ssh/id_rsa.pub -o StrictHostKeyChecking=no -o UserKnownHostsFile=/dev/null i-0812067ae399318e6
total 8
-rw-r--r-- 1 jenkins jenkins 1456 Jan 2 13:18 known_hosts
-rw-r--r-- 1 jenkins jenkins 823 Jan 1 18:42 known_hosts.old
[root@ip-172-31-24-55:~#]# ssh -i /root/.ssh/id_rsa.pub -o StrictHostKeyChecking=no -o UserKnownHostsFile=/dev/null i-0812067ae399318e6
total 8
-rw-r--r-- 1 jenkins jenkins 1456 Jan 2 13:18 known_hosts
-rw-r--r-- 1 jenkins jenkins 823 Jan 1 18:42 known_hosts.old
[root@ip-172-31-24-55:~#]# ssh -i /root/.ssh/id_rsa.pub -o StrictHostKeyChecking=no -o UserKnownHostsFile=/dev/null i-0812067ae399318e6
total 8
-rwxr-xr-x 1 jenkins jenkins 1456 Jan 2 13:18 known_hosts
-rw-r--r-- 1 jenkins jenkins 823 Jan 1 18:42 known_hosts.old
[root@ip-172-31-24-55:~#]#
```

i-0f506b3638890d1e4 (Master\_jenkins)  
Docker: 0.0.0.0:22 -> 172.17.0.1:22

### Create an agent

← → ↻ 🔒 Not secure 98.93.79.46:8081/manage/computer/new 🔍 ☆ 🗑️ 👤

📧 Gmail 📺 YouTube 📰 News 📍 Maps 🗨️ Translate 🌐 Google 🧠 Data Science Mind...

**Jenkins** / Manage Jenkins ▾ / Nodes ▾ / New node 🔍 ⚙️ 👤

**New node**

Node name

slave-01

Type

☐ Permanent Agent

Adds a plain, permanent agent to Jenkins. This is called "permanent" because Jenkins doesn't provide higher level of integration with these agents, such as dynamic provisioning. Select this type if no other agent types apply — for example such as when you are adding a physical computer, virtual machines managed outside Jenkins, etc.

Create

### Create a directory in slave which will be added in notes

```
/root
[root@ip-172-31-24-55:~]# mkdir slave-workspace
[root@ip-172-31-24-55:~]# pwd
/root
[root@ip-172-31-24-55:~]# cd slave-workspace/
[root@ip-172-31-24-55:slave-workspace]# pwd
/root/slave-workspace
[root@ip-172-31-24-55:slave-workspace]#
```

i-0812067ae399318e6 (Slave\_1\_instance)



Given following details



A screenshot of the Jenkins 'New Node' configuration page. The fields are filled with the following values:

- Name:** slave-01
- Description:** slave for java
- Number of executors:** 2
- Remote root directory:** /root/slave-workspace
- Labels:** java
- Usage:** Use this node as much as possible

Below the fields, there are links for 'Plain text' and 'Preview'.

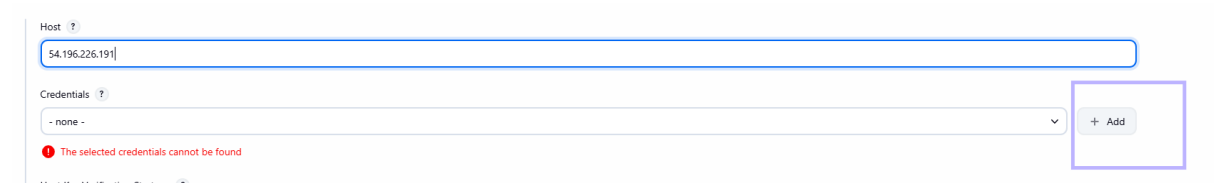
Select ssh add pem key and user will be ec2-user



A screenshot of the Jenkins 'Launch method' dropdown menu. The options are:

- Launch agent by connecting it to the controller
- Launch agent by connecting it to the controller
- Launch agents via SSH** (highlighted in blue)
- Keep this agent online as much as possible

Provide private host, slave public id



A screenshot of the Jenkins 'Host' and 'Credentials' fields. The 'Host' field contains the IP address 54.196.226.191. The 'Credentials' dropdown shows '- none -'. A red error message below the dropdown reads: 'The selected credentials cannot be found'. A purple box highlights the '+ Add' button next to the credentials dropdown.

Ssh username private key select this option

Domain

Global credentials (unrestricted)

Kind

Username with password

Username with password

GitHub App

SSH Username with private key

Secret file

Secret text

Certificate

Blank username: did you mean to use secret text credentials instead?

☐ Treat username as secret

Fill this following details

Jenkins Credentials Provider: Jenkins

ID

slave01-key

Description

Username

ec2-user

☐ Treat username as secret

Private Key

☒ Enter directly

Key

Enter New Secret Below

djLRBtP812rVbXPerIXuHCEG100Fhez08MPYcw12cnsRMBasQX0Pq8CgYEazpYo  
BBRmYqHTRy60MY26TYMut7+1N4mbttDqQ8107g34173NvdNDy3sC8QuqEck33Vr7  
/VktLrHJwfb1L1rmdYH102qWxK192FALBeVbFT4GNEs+58Psh2o1TYTAom1VWx  
L4hqW1TwuaIRNnSG8Bc1Xo19rr5k9cpGkoR2HsCgYB9G8UF9cd55/Ak224E9JtI  
vOX3mb0adbXSeDqW3CmNn8XIaDgkdLX7dORPm2FF2omuy8f9SEWJes3y3Pb1u1  
zrLhaDLZNTsxcHc1S1A4v31D4b8076g13RpyFxoXNSq29Yms53thkIn02rYqEU  
HpkPxeIEp/ASu8Kde1Pm3QK8gDyavF6GBYthy3aFswMeitskcvng4r8qN24ov7FQ  
XrPqhdFTZ003swYrY19Y8NfHb53BdeuqKOGHADDf8Cm1CH+QTYoFKPmPCCt

slave-01	N/A	N/A	N/A	N/A	N/A
Data obtained	2 ms	2 ms	1 ms	52 min	0 ms

Create a new job

New Item

Enter an item name

Java\_slave

Select an item type

☒ Freestyle project  
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.

☐ Maven project  
Build a maven project. Jenkins takes advantage of your POM files and drastically reduces the configuration.

☐ Pipeline  
Orchestrates long-running activities that can span multiple build agents. Suitable for building pipelines (formerly known as workflows) and/or organizing complex activities that do not easily fit in free-style job type.

☐ Multi-configuration project  
Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.

☐ Folder  
Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.

☐ Multibranch Pipeline  
Creates a set of Pipeline projects according to detected branches in one SCM repository.

OK

After creating a job we can select label in which we have created in node, or Agent

Validation:

Job is executed successfully.

📄 Status

Java\_slave

</> Changes

Permalinks

📁 Workspace

▶ Build Now

⚙️ Configure

🗑️ Delete Project

✎ Rename

📄 Add description

Builds > ⋮

🔍 Filter /

Today

✅ #1 2:35 PM ▼

